Topology and Geometry in Physics – Note 1

Ling-Wei Luo*

School of Fundamental Physics and Mathematical Sciences,
Hangzhou Institute for Advanced Study, UCAS

April 20, 2026

Abstract

This is the *first* note on the course of topology and geometry in physics at the School of Fundamental Physics and Mathematical Sciences, Hangzhou Institute for Advanced Study, UCAS.

Contents

*Electronic address: lwluo.phys@gmail.com

1 Affine space

Given a point set A and a vector space V. For any point $p \in A$ and a vector $\mathbf{v} \in V$, we can get a new point $q \in A$ through the action of a vector $\mathbf{v}$:

$$\begin{aligned} f : A \times V &\longrightarrow A, \\ (p, \mathbf{v}) &\longmapsto q := f(p, \mathbf{v}) = p + \mathbf{v}, \end{aligned} \tag{1.1}$$

with such kind of a structure we call a space an *affine space* denoted by a pair (A, V).

A vector space as an affine space. For every $p \in A$, we have an isomorphism from V to A:

$$\begin{aligned} f_p : V &\longrightarrow A, \\ \mathbf{v} &\longmapsto q = f_p(\mathbf{v}) = p + \mathbf{v}. \end{aligned} \tag{1.2}$$

Therefore, given a point $p \in A$ we can recognized the point p as a position vector $\mathbf{p} \in V = \mathbb{R}^n$ with respect to a reference point o as an *origin* in A by using the affine transformation $p = f_o(\mathbf{p}) = o + \mathbf{p}$. Let $\{e_i\}$ be a frame attached at o, the vector $\mathbf{p}$ can be written by $\mathbf{p} = e_i\, \xi^i$ with the coordinates $\{\xi^i\}$. Particularly $o = f_o(\mathbf{o}) = o + \mathbf{o}$ means that the point o can be represented by a zero vector $\mathbf{o} \in V$. Therefore A is isomorphic to a vector space V, and then we obtain an affine space (V, V) such that

$$\mathbf{p} = f_{\mathbf{o}}(\mathbf{p}) = \mathbf{o} + \mathbf{p} \quad \text{and} \quad \mathbf{q} = f_{\mathbf{p}}(\mathbf{v}) = \mathbf{p} + \mathbf{v} \tag{1.3}$$

with $\mathbf{v} \in V$, corresponding to

$$p = f_o(\mathbf{p}) = o + \mathbf{p} \quad \text{and} \quad q = f_p(\mathbf{v}) = p + \mathbf{v} \tag{1.4}$$

in (A, V), respectively. For $A = \mathbb{R}^2 = V$, affine spaces (A, V) and (V, V) are illustrated by Fig. 1.1.

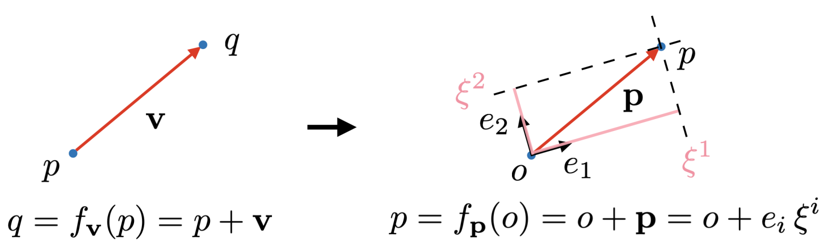

Fig. 1.1: Affine spaces (A, V) and (V, V) for $A = \mathbb{R}^2 = V$.

Traanslation. In addition, for every $\mathbf{v} \in V$, affine map induces a map of *translation* through a displacement vector $\mathbf{v}$ from the point p to point q:

$$\begin{aligned} f_{\mathbf{v}} : A &\longrightarrow A, \\ p &\longmapsto q = f_{\mathbf{v}}(p) = p + \mathbf{v}. \end{aligned} \tag{1.5}$$

With the structure of (V, V), the map becomes $f_{\mathbf{v}} : V \longrightarrow V$. There exist a displacement vector $\mathbf{v} = \mathbf{\Delta p}$ such that $\mathbf{q} = \mathbf{p} + \mathbf{\Delta p}$ corresponding to the translation $q = p + \mathbf{\Delta p}$, which is shown in Fig. 1.2.

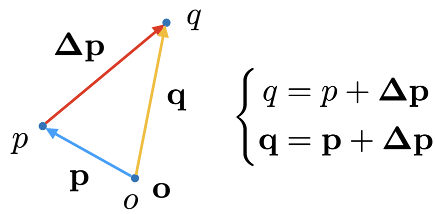

Fig. 1.2: Two points p and q joining by an infinitesimal displacement vector $\mathbf{\Delta p}$.

Parallel transport. Let $\mathbf{V}$ and $\mathbf{W}$ are two vectors. In affine space, we can arbitrarily parallel transport any vectors. The direction of a vector is always the same under the parallel transport. Under the parallel transport shown in Fig. 1.3, we can easily compare any two vectors at the

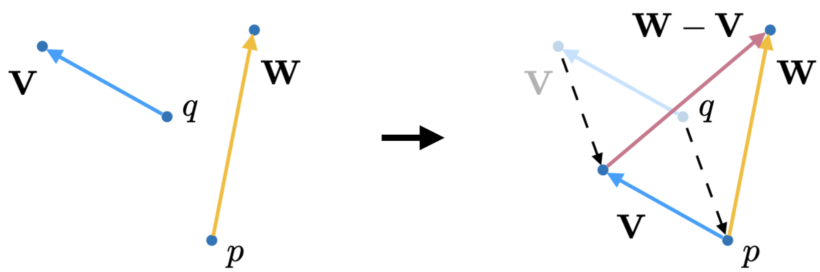

Fig. 1.3: Comparison of two vectors $\mathbf{V}$ and $\mathbf{W}$ at the sam point.

same point. If we compare a vector $\mathbf{V}$ to itself, it is apparent that it is always parallel to itself. Therefore, parallel transport gives the equation

$$\boxed{\mathbf{\Delta V} = 0,} \tag{1.6}$$

which is illustrated in Fig. 1.4.

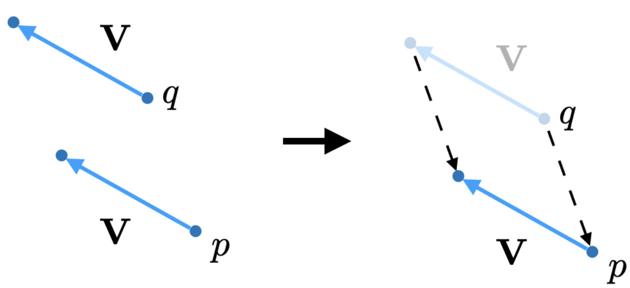

Fig. 1.4: A vector $\mathbf{V}$ is always parallel to itself.

2 Eclidean Space

Vector space. In a n-dimensional Euclidean space $\mathbb{R}^n$, we always label a point by coordinates $\{x^i\} = \{x_1, x_2, \ldots, x_n\}$. We can define a point as the origin o with coordinates $(x_1, x_2, \ldots, x_n) =$

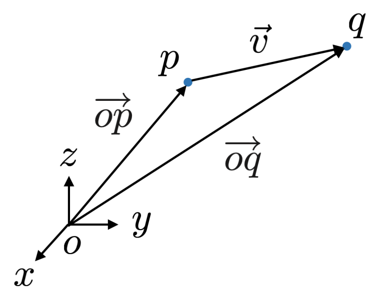

Fig. 2.1: A vector $\vec{v}$ in Euclidean space $\mathbb{R}^3$.

$(0, 0, \ldots, 0)$. In addition, any point can be regarded as an end point of an arrow pointed from the origin o. So Euclidean space also forms a *vector space*, which is a special *affine space*. We can naturally use the *Cartesian frame* $\{\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n\}$ as basis vector to describe such a directional arrow called *position vector*. Now we focus on the three-dimensional Euclidean space $\mathbb{R}^3$. Given two points p and q, shown in Fig. 2.1, and their coordinates are respectively labeled by (x, y, z) and (x', y', z'). We can have position vectors of points p and q written by $\overrightarrow{op} = x\,\vec{e}_x + y\,\vec{e}_y + z\,\vec{e}_z$ and $\overrightarrow{oq} = x'\,\vec{e}_x + y'\,\vec{e}_y + z'\,\vec{e}_z$ respectively.[†1] A *separation vector* $\vec{v}$ is always denoted by an arrow pointed from one point p to another point q. We assume that the separation vector between p and q is $\vec{v} \coloneqq \overrightarrow{pq}$. The separation vector $\vec{v}$ is an arrow pointed from p to q, which can be regarded as a vector represents the action of the movement of a point from p to q. We always call this movement the *displacement vector* at point p. Then, from linear algebra, the vector $\vec{v}$ is the difference of $\overrightarrow{oq}$ and $\overrightarrow{op}$ and we have the relation

$$\begin{aligned}
\vec{v} = \overrightarrow{pq} &= \underbrace{\overrightarrow{po}}_{-\overrightarrow{op}} + \overrightarrow{oq} = \overrightarrow{oq} - \overrightarrow{op} \\
&= (x' - x)\,\vec{e}_x + (y' - y)\,\vec{e}_y + (z' - z)\,\vec{e}_z \\
&\equiv v_x\,\vec{e}_x + v_y\,\vec{e}_y + v_z\,\vec{e}_z. \qquad (2.1)
\end{aligned}$$

Notations of a vector space. Generally we prefer to write a n-dimensional vector by

$$\vec{v} = \sum_i v_i\,\vec{e}_i \coloneqq v_i\,\vec{e}_i \overset{\text{or}}{=} \vec{e}_i\,v_i \quad (i = 1, 2, \ldots, n). \qquad (2.2)$$

Here we use the so-called *Einstein's summation convention* in which the repeated index should be realized by the summation over the all components. Consider a three-dimensional vector $\vec{v}$, there are two notations for a vector can be presented here:

(i) We always use the components (or coordinates) with a *parentheses* or *square bracket* to represent the vector $\vec{v}$ denoted by a *row matrix* (or a *tuple*)

$$\mathsf{v} \equiv (v_i) = \begin{pmatrix} v_x & v_y & v_z \end{pmatrix} \overset{\text{or}}{=} (v_x, v_y, v_z), \qquad (2.3\text{a})$$

or a *column matrix*

$$\mathsf{v} \equiv (v_i) = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}. \qquad (2.3\text{b})$$

[†1] We may sometimes recognize the points p and q by their position vectors denoted by $\mathbf{p} \equiv \overrightarrow{op}$ and $\mathbf{q} \equiv \overrightarrow{oq}$, respectively.

We note that *matrices* are always denoted by letters in $\mathsf{sans\ serif}$ typeface. Conventionally, people always refer a vector to a column matrix, and such a matrix expression of the vector $\vec{v}$, cf. (2.1), is

$$\vec{v} \quad \longrightarrow \quad \mathsf{v} \equiv (v_i) = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = v_x\,\mathsf{e}_x + v_y\,\mathsf{e}_y + v_z\,\mathsf{e}_z \tag{2.4}$$

by recognizing basis vectors as the following basis matrices

$$\vec{e}_x \equiv \mathsf{e}_x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \vec{e}_y \equiv \mathsf{e}_y = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{e}_z \equiv \mathsf{e}_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.^{\dagger 2} \tag{2.5}$$

(ii) We always prefer the expression $\vec{v} = \vec{e}_i\,v_i$ (basis times component). The basis vectors are represented by a *row* matrix, while the corresponding components are represented by a *column* matrix. So a vector can easily correspond to the multiplication of a row matrix (basis matrix) and a column matrix (component matrix). Define basis and component matrices by

$$\vec{e}_i \quad \longrightarrow \quad \mathsf{e} = (\mathsf{e}_i) = \begin{pmatrix} \vec{e}_1 & \vec{e}_2 & \cdots & \vec{e}_n \end{pmatrix}_{1\times n} \quad \text{and} \quad v_i \quad \longrightarrow \quad \mathsf{v} = (\mathsf{v}^i) = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}_{n\times 1}, \tag{2.6}$$

where i can also be standed for the index of a matrix element, and the upper and lower indices are represented the row and column indices respectively. Therefore, the vector $\vec{v}$ can be understood by the multiplication of matrices e and v:

$$\vec{v} = \vec{e}_i\,v_i \quad \longrightarrow \quad \vec{v} \equiv \mathsf{e}_i\mathsf{v}^i = \mathsf{e}\cdot\mathsf{v}, \quad \text{or more precisely} \quad \vec{v} \equiv \mathsf{e}_{1\times n}\cdot\mathsf{v}_{n\times 1}. \tag{2.7}$$

Inner product. A *bilinear operator* g in a vector space V is introduced such that

$$\text{linear in the first slot: } g(a\,\vec{x}_1 + b\,\vec{x}_2, \vec{y}_1) = a\,g(\vec{x}_1, \vec{y}_1) + b\,g(\vec{x}_2, \vec{y}_1), \tag{2.8a}$$

$$\text{linear in the second slot: } g(\vec{x}_1, c\,\vec{y}_1 + d\,\vec{y}_2) = c\,g(\vec{x}_1, \vec{y}_1) + d\,g(\vec{x}_1, \vec{y}_2), \tag{2.8b}$$

where $\vec{x}_1$, $\vec{x}_2$, $\vec{y}_1$ and $\vec{y}_2$ are vectors in V, and a, b, c and d are real constants. For simple case, we have the following equation

$$g(a\,\vec{x}, b\,\vec{y}\,) = ab\,g(\vec{x}, \vec{y}\,). \tag{2.9}$$

We say that the inner product of two vectors is a bilinear operation of g which can be identified as a function of two *vectors*

$$g(\vec{v}, \vec{w}) = g(\vec{e}_i\,v_i, \vec{e}_j\,w_j) = v_i\,g(\vec{e}_i, \vec{e}_j)\,w_j. \tag{2.10}$$

We can define a matrix

$$\mathsf{G} = (\mathsf{G}^i{}_j) \equiv (g_{ij}), \tag{2.11}$$

[†2] This notation is always used as the notation of a *basis ket vector* in *quantum mechanics*.

where

$$g_{ij} \coloneqq g(\vec{e}_i, \vec{e}_j) = \vec{e}_i \cdot \vec{e}_j = \|\vec{e}_i\| \, \|\vec{e}_j\| \, \cos\theta_{ij}. \tag{2.12}$$

If $\{e_i\}$ is a *normalized* frame, then we would have

$$g_{ij} = \cos\theta_{ij}. \tag{2.13}$$

where θ_{ij} is the angle between $\vec{e}_i$ and $\vec{e}_j$. Consequently, g_{ij} given in a *normalized* frame is the *direction cosine* of the ith axis relative to the jth axis. Hence, the inner product can be expressed by a multiplication of matrix forms

$$g(\vec{v}, \vec{w}) = v_i \, g_{ij} \, w_j \equiv \mathsf{v}_i \, \mathsf{G}^i{}_j \, \mathsf{w}^j = \mathsf{v}^{\mathrm{T}} \cdot \mathsf{G} \cdot \mathsf{w}, \tag{2.14}$$

where T stands for the *transpose* operation of a matrix and

$$\mathsf{v}^{\mathrm{T}} = (\mathsf{v}_i) \equiv \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix}, \quad \mathsf{G} = (\mathsf{G}^i{}_j) \equiv \begin{pmatrix} g_{11} & g_{12} & \cdots & g_{1n} \\ g_{21} & g_{22} & \cdots & g_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ g_{n1} & g_{n2} & \cdots & g_{nn} \end{pmatrix}, \quad \mathsf{w} = (\mathsf{w}^j) \equiv \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}. \tag{2.15}$$

If $\{\vec{e}_i\}$ is an *orthonormal* frame, then the matrix $g_{ij} = \delta_{ij}$, where

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \longrightarrow \quad \mathsf{G} = \mathrm{diag}\,(\underbrace{1, 1, \cdots, 1}_{n \text{ terms}}) = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix}_{n\times n} \tag{2.16}$$

is called the *Kronecker delta symbol*. In this situation, we can rewrite the inner product

$$g(\vec{v}, \vec{w}) = v_i \, \delta_{ij} \, w_j \equiv \mathsf{v}^{\mathrm{T}} \cdot \mathsf{G} \cdot \mathsf{w} \tag{2.17a}$$

$$= \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix} \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \tag{2.17b}$$

$$= \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \tag{2.17c}$$

$$= v_1 w_1 + v_2 w_2 + \cdots + v_n w_n, \tag{2.17d}$$

which is consist with the ordinary inner product of $\vec{v}$ and $\vec{w}$ in the Euclidean space, i.e., we obtain the identification of

$$\boxed{g(\vec{v}, \vec{w}) = v_i \, g_{ij} \, w_j = \vec{v} \cdot \vec{w},} \tag{2.18}$$

which leads to the norm of a vector $\vec{v}$ by

$$\boxed{\|\vec{v}\| = \sqrt{(v^1)^2 + (v^2)^2 + (v^3)^2}.} \tag{2.19}$$

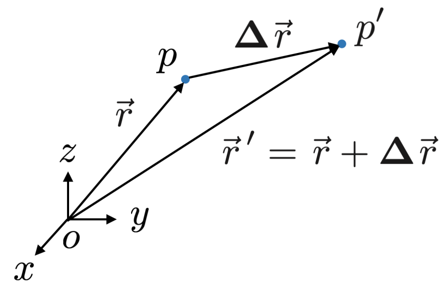

Fig. 2.2: A displacement vector $\boldsymbol{d}\vec{r}$ form the point p to the point q.

The line element. In $\mathbb{R}^3$, we naturally have the position vectors of points p and $p' \equiv q$ are respectively written as $\vec{r} = \vec{e}_x\, x + \vec{e}_y\, y + \vec{e}_z\, z$ and $\vec{r}\,' = \vec{r} + \boldsymbol{\Delta}\vec{r} = \vec{e}_x\, x' + \vec{e}_y\, y' + \vec{e}_z\, z'$, We note that $\boldsymbol{\Delta}\vec{r}$ is a displacement vector from p to q.[†3] This is shown in Fig. 2.2. To emphasize the *fixed* Cartesian frame, we sometime denote it by the notation of $\{\vec{e}_i\} = \{\vec{\delta}_i\}$ for $i = x, y, z$. Thus the displacement vector in *Cartesian coordinates* is usually read by

$$\begin{aligned} \boldsymbol{\Delta}\vec{r} &= \overrightarrow{pq} = \overrightarrow{oq} - \overrightarrow{op} = \vec{r}\,' - \vec{r} && \text{(2.20a)}\\ &= \vec{e}_x\, \Delta x + \vec{e}_y\, \Delta y + \vec{e}_z\, \Delta z \quad (\overset{\text{or}}{=} \vec{\delta}_i\, \Delta x^i). && \text{(2.20b)} \end{aligned}$$

We note that $\boldsymbol{\Delta}$ in **bold-face** is used to denote the *variation of a vector* (a *vectorial* variation) in order to distinguish the ordinary variation of a function, which includes not only the difference of a function but also the addition law of vectors in linear algebra. The variation of a displacement vector is realized by

$$\begin{aligned} \boldsymbol{\Delta}\vec{r} &= \boldsymbol{\Delta}(\vec{e}_x\, x + \vec{e}_y\, y + \vec{e}_z\, z)\\ &= \underbrace{\boldsymbol{\Delta}\vec{e}_x}_{0}\, x + \vec{e}_x\, \Delta x + \underbrace{\boldsymbol{\Delta}\vec{e}_y}_{0}\, y + \vec{e}_y\, \Delta y + \underbrace{\boldsymbol{\Delta}\vec{e}_z}_{0}\, z + \vec{e}_z\, \Delta z. \end{aligned} \tag{2.21}$$

Therefore, due to the *invariant* basis of the Cartesian frame $\{\vec{e}_i\}$, the result obtained coincides with (2.20). The norm of the displacement vector $\boldsymbol{\Delta}\vec{r}$ is denoted by ΔS, which can be obtained by the inner product:

$$\Delta S := \|\boldsymbol{\Delta}\vec{r}\| = \sqrt{\boldsymbol{\Delta}\vec{r}\cdot\boldsymbol{\Delta}\vec{r}} = \sqrt{g(\boldsymbol{\Delta}\vec{r}, \boldsymbol{\Delta}\vec{r})}. \tag{2.22}$$

We define the *line element* of a space as the square of the norm of $\boldsymbol{\Delta}\vec{r}$ by

$$\Delta S^2 = \boldsymbol{\Delta}\vec{r}\cdot\boldsymbol{\Delta}\vec{r} = g\big(\boldsymbol{\Delta}\vec{r}, \boldsymbol{\Delta}\vec{r}\big) = g\big(\vec{e}_i\, \Delta x_i, \vec{e}_j\, \Delta x_j\big) = g\big(\vec{e}_i, \vec{e}_j\big)\Delta x_i \Delta x_j = g_{ij}\, \Delta x_i \Delta x_j. \tag{2.23}$$

In physics people are always interested in the *infinitesimal* form. If two points p and q are separated by an *infinitesimal distance* dS, then we have the *vectorial differential* of the position vector

$$\begin{aligned} \boldsymbol{\Delta}\vec{r} \quad \longrightarrow \quad \boldsymbol{d}\vec{r} &= \overrightarrow{pq} = \overrightarrow{oq} - \overrightarrow{op} = \vec{r}\,' - \vec{r} && \text{(2.24a)}\\ &= \vec{e}_x\, dx + \vec{e}_y\, dy + \vec{e}_z\, dz \quad (\overset{\text{or}}{=} \delta_i\, dx_i). && \text{(2.24b)} \end{aligned}$$

[†3] Generally we can denote the position vector of the point p by $\mathbf{p}$ (or by a matrix expression p) and write the displacement vector by $\boldsymbol{\Delta}\mathbf{p} \equiv \mathbf{p}' - \mathbf{p}$ (or by a matrix expression $\boldsymbol{\Delta}\mathsf{p}$) or simply $\boldsymbol{\Delta}p$. Correspondingly, its infinitesimal form should be $\boldsymbol{d}\mathbf{p}$ (or by a matrix expression $\boldsymbol{d}\mathsf{p}$) or simply $\boldsymbol{d}p$ respectively.

So the *line element* of a space becomes

$$dS^2 = \boldsymbol{d}\vec{r} \cdot \boldsymbol{d}\vec{r} = g\big(\boldsymbol{d}\vec{r}, \boldsymbol{d}\vec{r}\big) = g_{ij}\, dx_i dx_j. \tag{2.25}$$

For two-dimensional Euclidean space, the displacement vector is $\boldsymbol{d}\vec{r} = \vec{e}_x\, dx + \vec{e}_y\, dy = \vec{e}_i\, dx_i$ and the corresponding line element with $g_{ij} = \delta_{ij}$ is

$$\begin{aligned} dS^2 &= \delta_{ij}\, dx_i dx_j &(2.26\text{a})\\ &\equiv \begin{pmatrix} dx & dy \end{pmatrix} \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix}. &(2.26\text{b}) \end{aligned}$$

Similarly, the three-dimensional line element with $g_{ij} = \delta_{ij}$ and $\boldsymbol{d}\vec{r} = \vec{e}_x\, dx + \vec{e}_y\, dy + \vec{e}_z\, dz = \vec{e}_i\, dx_i$ is

$$\begin{aligned} dS^2 &= \delta_{ij}\, dx_i dx_j &(2.27\text{a})\\ &\equiv \begin{pmatrix} dx & dy & dz \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix}. &(2.27\text{b}) \end{aligned}$$

Example. (Curvilinear Coordinates).

In stead of using the Cartesian coordinates, we can choose *curvilinear coordinates* for calculations. We would like to know the displacement vector $\boldsymbol{d}\vec{r} = \vec{r}\,' - \vec{r}$ and line element in these coordinate systems.

Polar coordinates in $\mathbb{R}^2$.

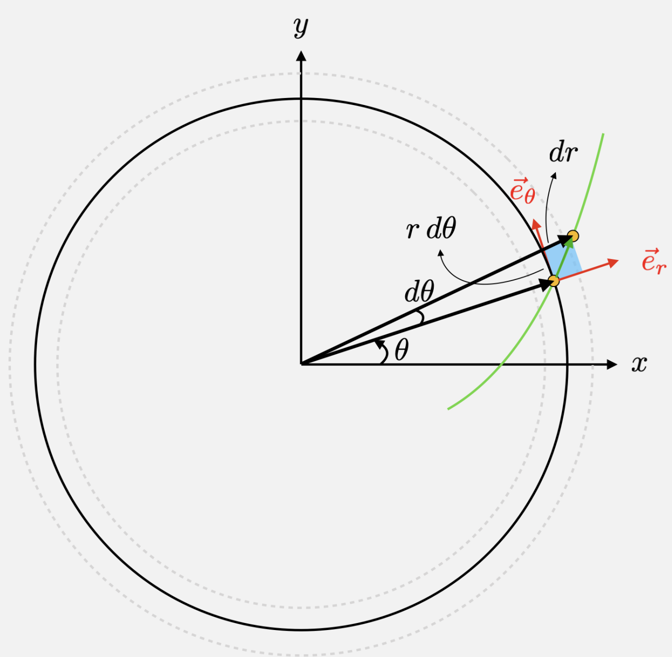

Fig. 2.3: Polar coordinates in $\mathbb{R}^2$.

Given a position vector

$$\vec{r}(x,y) = \vec{e}_x\, x + \vec{e}_y\, y, \tag{2.28}$$

where the coordinates (x, y) in terms of the polar coordinates are

$$\begin{cases} x = r\cos\theta, & (2.29\text{a}) \\ y = r\sin\theta. & (2.29\text{b}) \end{cases}$$

So we would like to consider the coordinate transformation

$$(x,y) \quad \longrightarrow \quad (r,\theta). \tag{2.30}$$

What about the coordinate transformation of the basis vector? That means we will have

$$\{\vec{e}_x, \vec{e}_y\} \quad \longrightarrow \quad \{\vec{e}_r, \vec{e}_\theta\}. \tag{2.31}$$

We know that basis vectors should be the tangent vectors along the corresponding direction of coordinates. Hence, we can obtain basis vectors through the derivatives with respect to the orresponding coordinates

$$\begin{aligned} \vec{V}_r = \frac{\partial \vec{r}}{\partial r} &= \frac{\partial(\vec{e}_x\, x + \vec{e}_y\, y)}{\partial r} \\ &= \underbrace{\frac{\partial \vec{e}_x}{\partial r}}_{0}\, r\cos\theta + \vec{e}_x\, \frac{\partial(r\cos\theta)}{\partial r} + \underbrace{\frac{\partial \vec{e}_y}{\partial r}}_{0}\, r\sin\theta + \vec{e}_y\, \frac{\partial(r\sin\theta)}{\partial r} \\ &= \vec{e}_x \cos\theta + \vec{e}_y \sin\theta \quad \longrightarrow \quad \boxed{\vec{e}_r := \frac{\vec{V}_r}{\|\vec{V}_r\|} = \vec{e}_x\cos\theta + \vec{e}_y\sin\theta,} \qquad (2.32\text{a}) \\ \vec{V}_\theta = \frac{\partial \vec{r}}{\partial \theta} &= \frac{\partial(\vec{e}_x\, x + \vec{e}_y\, y)}{\partial r} \\ &= \vec{e}_x\, \frac{\partial(r\cos\theta)}{\partial\theta} + \vec{e}_y\, \frac{\partial(r\sin\theta)}{\partial\theta} \\ &= -\vec{e}_x\, r\sin\theta + \vec{e}_y\, r\cos\theta \quad \longrightarrow \quad \boxed{\vec{e}_\theta := \frac{\vec{V}_\theta}{\|\vec{V}_\theta\|} = -\vec{e}_x\sin\theta + \vec{e}_y\cos\theta,} \qquad (2.32\text{b}) \end{aligned}$$

where the *scale factors* are defined as the *norm* of tangent vectors

$$\|\vec{V}_r\| = 1 := h_r, \tag{2.33a}$$

$$\|\vec{V}_\theta\| = r := h_\theta. \tag{2.33b}$$

The position vector in the polar coordinates is

$$\boxed{\begin{array}{ll} & \vec{r} = \vec{e}_x(r\cos\theta) + \vec{e}_y(r\sin\theta) = \vec{e}_r\, r \\ \Longrightarrow & \text{No cooordinate } \theta \text{ in the position vector.} \\ \Longrightarrow & \text{We would have a } \textit{symmetry} \text{ in the } \theta\text{-direction.} \end{array}} \tag{2.34}$$

By differentiating the position vector, the displacement vector can be regarded as a total differential of the vector $\vec{r}$:

$$\begin{aligned}
\boldsymbol{d}\vec{r} &= \boldsymbol{d}\big(\vec{e}_x\, x + \vec{e}_y\, y\big) \\
&= \underbrace{\boldsymbol{d}\vec{e}_x}_{0}(r\,\cos\theta) + \vec{e}_x\,\boldsymbol{d}(r\,\cos\theta) + \underbrace{\boldsymbol{d}\vec{e}_y}_{0}(r\,\sin\theta) + \vec{e}_y\,\boldsymbol{d}(r\,\sin\theta) \\
&= \vec{e}_x(dr\,\cos\theta - r\,\sin\theta\, d\theta) + \vec{e}_y(dr\,\sin\theta + r\,\cos\theta\, d\theta) \\
&= \underbrace{(\vec{e}_x\,\cos\theta + \vec{e}_y\,\sin\theta)}_{\vec{e}_r\, h_r = \vec{V}_r = \frac{\partial\vec{r}}{\partial r}} dr + \underbrace{(-\vec{e}_x\,\sin\theta + \vec{e}_y\,\cos\theta)\, r}_{\vec{e}_\theta\, h_\theta = \vec{V}_\theta = \frac{\partial\vec{r}}{\partial\theta}} d\theta.
\end{aligned} \tag{2.35}$$

On the other hand, the displacement vector can be directly realized in the polar coordinates by

$$\boldsymbol{d}\vec{r} = \boldsymbol{d}(\vec{e}_r\, r) = \boldsymbol{d}\vec{e}_r + \vec{e}_r\, dr. \tag{2.36}$$

By comparing (2.35) and (2.36), we have found that

$$\boxed{\boldsymbol{d}\vec{e}_r = \vec{e}_\theta\, d\theta.} \tag{2.37}$$

Alternatively, we can use (2.32) respectively to calculate $\boldsymbol{d}\vec{e}_r$ and $\boldsymbol{d}\vec{e}_\theta$:

$$\begin{cases}
\boldsymbol{d}\vec{e}_r = -\vec{e}_x \sin\theta\, d\theta + \vec{e}_y \cos\theta\, d\theta = +\vec{e}_\theta\, d\theta, & (2.38\text{a}) \\
\boldsymbol{d}\vec{e}_\theta = -\vec{e}_x \cos\theta\, d\theta - \vec{e}_y \sin\theta\, d\theta = -\vec{e}_r\, d\theta. & (2.38\text{b})
\end{cases}$$

Hence, it results in that the displacement vector in the polar coordinates is given by

$$\boxed{\boldsymbol{d}\vec{r} = \frac{\partial\vec{r}}{\partial r}\, dr + \frac{\partial\vec{r}}{\partial\theta}\, d\theta = \vec{V}_r\, dr + \vec{V}_\theta\, d\theta \equiv \vec{e}_r(h_r\, dr) + \vec{e}_\theta(h_\theta\, d\theta),} \tag{2.39}$$

the sets $\{\vec{V}_r, \vec{V}_\theta\}$ and $\{\vec{e}_r, \vec{e}_\theta\}$ are both basis vector of the tangent space. The *line element* in coordinates (r, θ) is defined as the square of the norm of the displacement vector $\boldsymbol{d}\vec{r}$:

$$\begin{aligned}
dS^2 = g(\boldsymbol{d}\vec{r}, \boldsymbol{d}\vec{r}) &= h_r^2\, dr^2 + h_\theta^2\, d\theta^2 \\
&\equiv \begin{pmatrix} h_r\, d\rho & h_\theta\, d\theta \end{pmatrix} \begin{pmatrix} h_r\, dr \\ h_\theta\, d\theta \end{pmatrix} \\
&= \begin{pmatrix} h_r\, d\rho & h_\theta\, d\theta \end{pmatrix} \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \begin{pmatrix} h_r\, dr \\ h_\theta\, d\theta \end{pmatrix} \\
&\overset{\text{or}}{\equiv} \begin{pmatrix} d\rho & d\theta \end{pmatrix} \underbrace{\begin{pmatrix} h_r^2 & \\ & h_\theta^2 \end{pmatrix}}_{g_{ij}} \begin{pmatrix} dr \\ d\theta \end{pmatrix}
\end{aligned} \tag{2.40}$$

with the metric tensor is

$$g_{ij} = \vec{V}_i \cdot \vec{V}_j = \begin{pmatrix} \vec{V}_r \cdot \vec{V}_r & \\ & \vec{V}_\theta \cdot \vec{V}_\theta \end{pmatrix} \equiv \begin{pmatrix} h_r^2 & \\ & h_\theta^2 \end{pmatrix} = \begin{pmatrix} 1 & \\ & r^2 \end{pmatrix} \quad (i, j = r, \theta). \tag{2.41}$$

In most situations, we would like to know the motion of a particle described, i.e., the trajectory of a particle, which is represented by the position vector $\vec{r}$. We needs to solve the equation of motion to and then obtain the solution $\vec{r}$. Therefore, it is hard to know the symmetry by reading the information from $\vec{r}$ initially, such as what we did in (2.34). However, the *displacement vector*(2.39) or the *line element* (2.40) contains most information of the geometric structure, we can start to calculate what we want, e.g., the equation of motion, we will explain in the following discussions, from one of them, and finally find the solution of $\vec{r}$.

Cylindrical coordinates in $\mathbb{R}^3$.

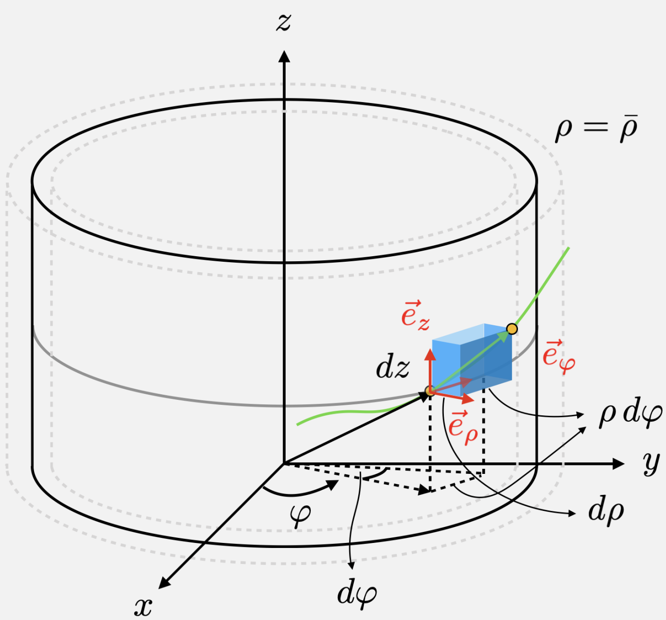

Fig. 2.4: Cylindrical coordinates in $\mathbb{R}^3$.

The coordinates (x, y, z) in terms of the cylindrical coordinates are

$$\begin{cases} x = \rho \cos\varphi, & (2.42a) \\ y = \rho \sin\varphi, & (2.42b) \\ z = z. & (2.42c) \end{cases}$$

Hence, the basis vectors should be

$$\begin{cases} \vec{e}_\rho = \vec{e}_x \cos\varphi + \vec{e}_y \sin\varphi, & (2.43a) \\ \vec{e}_\varphi = -\vec{e}_x \sin\varphi + \vec{e}_y \cos\varphi, & (2.43b) \\ \vec{e}_z = \vec{e}_z. & (2.43c) \end{cases}$$

The position vector is

$$\boxed{\begin{aligned} & \vec{r} = \vec{\rho} + \vec{z} = \vec{e}_\rho\, \rho + \vec{e}_z\, z \\ \Longrightarrow\quad & \text{The } \varphi\text{-direction provides symmetry.} \end{aligned}}$$

The displacement vector

$$\begin{aligned} \boldsymbol{d}\vec{r} &= \boldsymbol{d}(\vec{e}_\rho\,\rho + \vec{e}_z\,z) = \underbrace{\boldsymbol{d}\vec{e}_\rho}_{\vec{e}_\varphi\,d\varphi}\,\rho + \vec{e}_\rho\,d\rho + \underbrace{\boldsymbol{d}\vec{e}_z}_{0}\,z + \vec{e}_z\,dz \\ &= \vec{e}_\rho\,d\rho + \vec{e}_\varphi(\rho\,d\varphi) + \vec{e}_z\,dz. \end{aligned} \tag{2.44}$$

The line element in (ρ, φ, z) can be written by

$$\begin{aligned} dS^2 &= d\rho^2 + \rho^2\,d\varphi^2 + dz^2 \\ &\equiv \begin{pmatrix} d\rho & d\varphi & dz \end{pmatrix} \underbrace{\begin{pmatrix} 1 & & \\ & \rho^2 & \\ & & 1 \end{pmatrix}}_{g_{ij}} \begin{pmatrix} d\rho \\ d\varphi \\ dz \end{pmatrix}. \end{aligned} \tag{2.45}$$

Spherical coordinates in $\mathbb{R}^3$.

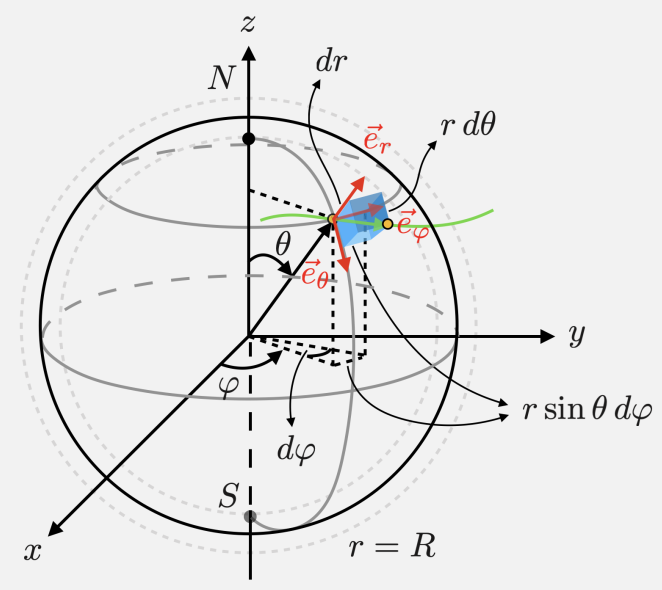

Fig. 2.5: Spherical coordinates in $\mathbb{R}^3$.

The coordinates (x, y, z) in terms of the spherical coordinates are

$$\begin{cases} x = r\sin\theta\cos\varphi, & (2.46a) \\ y = r\sin\theta\sin\varphi, & (2.46b) \\ z = r\cos\theta. & (2.46c) \end{cases}$$

Hence, the basis vectors should be

$$\begin{cases} \vec{e}_r = \vec{e}_x\sin\theta\cos\varphi + \vec{e}_y\sin\theta\sin\varphi + \vec{e}_z\cos\theta, & (2.47a) \\ \vec{e}_\theta = \vec{e}_x\cos\theta\cos\varphi + \vec{e}_y\cos\theta\sin\varphi - \vec{e}_z\sin\theta, & (2.47b) \\ \vec{e}_\varphi = -\vec{e}_x\sin\varphi + \vec{e}_y\cos\varphi. & (2.47c) \end{cases}$$

The position vector in the spherical coordinates is

$$\boxed{\begin{array}{ll} & \vec{r} = \vec{e}_r\, r \\ \Longrightarrow & \text{There exists a symmetry on the } (\theta, \varphi)\text{-surface.} \end{array}}$$

The displacement vector is

$$\boldsymbol{d}\vec{r} = \boldsymbol{d}(\vec{e}_r\, r) = \underbrace{\boldsymbol{d}\vec{e}_r}_{\vec{e}_\theta\, d\theta + \vec{e}_\varphi(\sin\theta\, d\varphi)} r + \vec{e}_r\, dr = \vec{e}_r\, dr + \vec{e}_\theta(r\, d\theta) + \vec{e}_\varphi(r\, \sin\theta\, d\varphi). \tag{2.48}$$

The line element in (r, θ, φ) can be written by

$$\begin{aligned} dS^2 &= dr^2 + r^2\, d\theta^2 + r^2\, \sin^2\theta\, d\varphi^2 \\ &\equiv \begin{pmatrix} dr & d\theta & d\varphi \end{pmatrix} \underbrace{\begin{pmatrix} 1 & & \\ & r^2 & \\ & & r^2 \sin^2\theta \end{pmatrix}}_{g_{ij}} \begin{pmatrix} dr \\ d\theta \\ d\varphi \end{pmatrix}. \end{aligned} \tag{2.49}$$

3 Classical Differential Geometry

3.1 Curves on $\mathbb{R}^2$

Consider a 2-dimensional Euclidean space, which is an $\mathbb{R}^2$ plane, which is constructed by a *fixed* frame. So any point in this plane can be described by a set coordinate function with respect to a reference point o, a 2-dimensional *Cartesian coordinate system*. A curve can be realized by a time

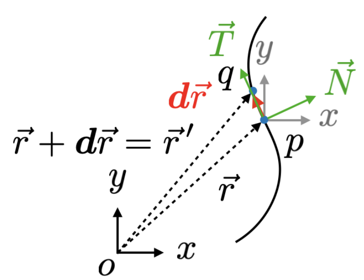

Fig. 3.1: A curve on $\mathbb{R}^2$.

series set of points, and hence, we parameterize a curve as a vector-valued function of one variable $t \in \mathbb{R}$, i.e., $\vec{r}(t) = \big(x(t), y(t)\big)^T$. The *velocity* vector at point $\vec{r}$ is given by

$$\vec{v}(t) = \dot{\vec{r}}(t) := \frac{\boldsymbol{d}\vec{r}}{dt} = e_x \frac{dx}{dt} + e_y \frac{dy}{dt} = \big(\dot{x}(t), \dot{y}(t)\big)^T, \tag{3.1}$$

where $\boldsymbol{d}\vec{r} = e_x\, dx + e_y\, dy + e_z\, dz$ and the *norm* of $\vec{v}$ is

$$\|\vec{v}\,\| = \|\dot{\vec{r}}(t)\| = \sqrt{\dot{\vec{r}} \cdot \dot{\vec{r}}} = \sqrt{\dot{x}^2 + \dot{y}^2}\,, \tag{3.2}$$

where $\dot{x} \coloneqq dx/dt$. Then the arc length between a interval $[a, t]$ can be regarded as a function of parameter t given by

$$s = \int_a^t \|\dot{\vec{r}}(t')\| \, dt' \equiv f(t). \tag{3.3}$$

It can be shown that by using s as a parameter of a curve $\vec{r}(t(s))$ with $t = f^{-1}(s)$, the norm of the velocity is normalized, i.e.,

$$\|\vec{r}\,'(s)\| = \sqrt{x'^2 + y'^2} = \sqrt{\dot{x}^2 + \dot{y}^2}\,\frac{dt}{ds} = \|\dot{\vec{r}}\,\|\,\frac{dt}{ds} = \frac{df}{dt}\frac{dt}{ds} = 1, \tag{3.4}$$

where $x' \coloneqq dx/ds$. Therefore, we can define a *unit tangent vector* as a *velocity* vector parameterized by s

$$\boxed{\vec{T} \coloneqq \vec{r}\,'(s).} \tag{3.5}$$

By differentiate $\vec{r}\,' \cdot \vec{r}\,' = 1$ with respect to s again, we obtain

$$\vec{r}\,'' \cdot \vec{r}\,' + \vec{r}\,' \cdot \vec{r}\,'' = 0 \qquad \Longrightarrow \qquad \vec{r}\,'' \cdot \vec{r}\,' = 0, \tag{3.6}$$

so the *acceleration* vector $\vec{a}(s) \coloneqq \vec{r}\,'' = \vec{T}\,'$ is normal to $\vec{T} = \vec{r}\,'$.

> For any *unit* vector $\vec{V}$ with the norm $\|\vec{V}\| = 1$, the corresponding derivative vector would be perpendicular to itself, i.e.,
>
> $$\vec{V}' \perp \vec{V}\,. \tag{3.7}$$

Due to the orthonormality, we can define a *principal normal vector*

$$\boxed{\vec{N} = \frac{\vec{r}\,''}{\|\vec{r}\,''\|},} \tag{3.8}$$

which is a *unit* normal vector. We can rewrite (3.8) as

$$\boxed{\vec{T}\,' = \kappa(s)\,\vec{N}} \tag{3.9}$$

related to the normal vector $\vec{N}$, which spans a space normal to the tangent space spanned by $\vec{T}$ at any point of the curve $\vec{r}(s)$. The factor

$$\boxed{\kappa(s) \coloneqq \|\vec{r}\,''\| = \|\vec{a}(s)\|} > 0 \tag{3.10}$$

as the *magnitude* of the acceleration vector is defined as the *curvature* of the curve $\vec{r}(s)$ at s.

> Kinematically, the curvature of a space can be realized by the motion of a particle:
>
> $$\text{Curvature} \sim \text{Acceleration}. \tag{3.11}$$

Now we can use $\vec{T}$ and $\vec{N}$ as a frame $(\vec{r}; \vec{T}, \vec{N})$ to span a two-dimensional space which is called the *osculating plane*.

Since $\vec{N}'$ can be uniquely determined by $\vec{T}$ due to $\vec{N}' \cdot \vec{N} = 0$, let $\vec{N}' = a\vec{T}$. By $\vec{T} \cdot \vec{N} = 0$, we can differentiate the relation and obtain the solution of a:

$$0 = \vec{T}' \cdot \vec{N} + \vec{T} \cdot \vec{N}' = \kappa \vec{N} \cdot \vec{N} + \vec{T} \cdot \vec{N}' \quad \Longrightarrow \quad a = \vec{T} \cdot \vec{N}' = -\kappa \quad \text{or} \quad \boxed{\vec{N}' = -\kappa \vec{T}.} \tag{3.12}$$

As a result, we have the derivation equations of a frame $(\vec{r}; \vec{T}, \vec{N})$:

$$\begin{cases} \vec{r}\,' = \ +\vec{T}, \\ \vec{T}\,' = \qquad\quad +\kappa \vec{N}, \\ \vec{N}' = -\kappa \vec{T}, \end{cases} \quad \Longrightarrow \quad \begin{pmatrix} \vec{r}\,' \\ \vec{T}\,' \\ \vec{N}' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \kappa \\ -\kappa & 0 \end{pmatrix} \begin{pmatrix} \vec{T} \\ \vec{N} \end{pmatrix}, \tag{3.13}$$

which is called the *Frenet-Serret formula* in $\mathbb{R}^2$.

Example (A circle in $\mathbb{E}^2$).

A circle with radius r can be parametrized by $\vec{r}(t) = (r\cos t, r\sin t)$ with $0 \le t \le 2\pi$.

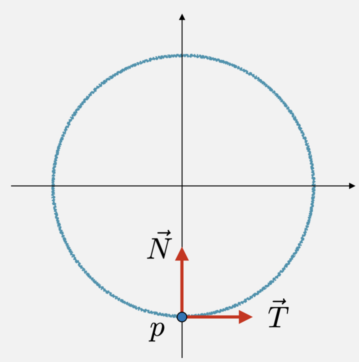

Fig. 3.2: A circle.

The tangent vector is

$$\dot{\vec{r}}(t) = (-r\sin t, r\cos t) \tag{3.14}$$

with norm

$$\|\dot{\vec{r}}\| = \sqrt{r^2 \sin^2 t, r^2 \cos^2 t)} := r\,. \tag{3.15}$$

The arc length $s(t)$ is

$$s(t) = \int_0^t \|\vec{r}(t')\| dt' = \int_0^t r\, dt' = rt\,. \tag{3.16}$$

Therefore, the circumference is

$$L = \int_0^{2\pi} \|\vec{r}(t')\| dt' = \int_0^{2\pi} r\, dt' = 2\pi r\,. \tag{3.17}$$

By $t = s/r$, the circle $\vec{r}(s)$ and its tangent vector are

$$\vec{r}(s) = \left(r\cos\frac{s}{r}, r\sin\frac{s}{r}\right) \tag{3.18a}$$

and

$$\vec{r}\,'(s) = \left(-\sin\frac{s}{r}, \cos\frac{s}{r}\right) = \vec{T} \tag{3.18b}$$

respectively. From (3.18b), we have

$$\vec{T}\,'(s) = \left(-\frac{1}{r}\cos\frac{s}{r}, -\frac{1}{r}\sin\frac{s}{r}\right). \tag{3.19}$$

The curvature κ can be obtained by

$$\kappa = \|\vec{T}\,'\| = \sqrt{\frac{1}{r^2}\cos^2\frac{s}{r} + \frac{1}{r^2}\sin^2\frac{s}{r}} = \frac{1}{r}\,, \tag{3.20}$$

which is the inverse of the constant radius r. The normal vector can be calculated by

$$\vec{N} = \frac{\vec{T}\,'}{\|\vec{T}\,'\|} = r\left(-\frac{1}{r}\cos\frac{s}{r} - \frac{1}{r}\sin\frac{s}{r}\right) = \left(-\cos\frac{s}{r}, \sin\frac{s}{r}\right). \tag{3.21}$$

3.2 Gauss Map

Gauss map G of a curve is a mapping which globally send all the points p of a curve to an *unit circle* S^1 (a Gauss circle) centered at C and send the corresponding normal vector $\vec{N}$ to a radius vector from C pointing to S^1, which is shown as Fig. 3.3. Therefore, $\vec{N}$ can be represented as a position vector of a point on S^1.

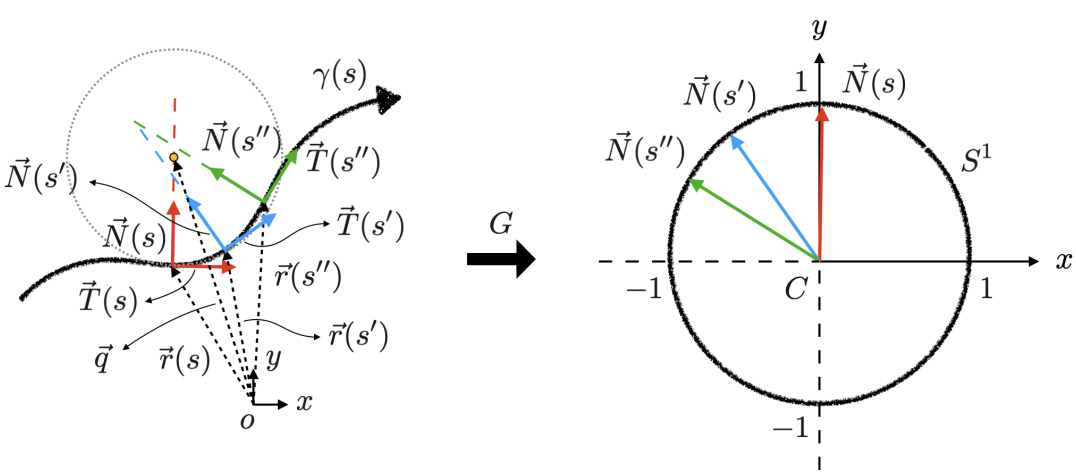

Fig. 3.3: The Gauss map G of a curve.

It is interesting to find a correspondence between the curve γ and the Gauss circle S^1:

$$\vec{r} \text{ of a curve } \gamma \;\sim\; \vec{N} \text{ of } S^1 \quad \Longrightarrow \quad \boldsymbol{d}\vec{r} \text{ of a curve } \gamma \;\sim\; \boldsymbol{d}\vec{N} \text{ of } S^1. \tag{3.22}$$

Let's consider two normal vectors $\vec{N}(s)$ and $\vec{N}(s')$ with respect to two infinitesimally nearby points $\vec{r}(s)$ and $\vec{r}(s')$, where $s' = s + ds$ is infinitesimally close to s. We can expand $\vec{N}(s')$ at s:

$$\begin{aligned} \vec{N}(s') &= \vec{N}(s + ds) \\ &\approx \vec{N}(s) + \vec{N}'(s)ds \\ &= \vec{N}(s) + \big(-\kappa(s)\vec{T}(s)\big)ds \\ &= \vec{N}(s) + \big(-\kappa(s)\,ds\big)\vec{T}(s)\,, \end{aligned} \tag{3.23}$$

which is the parametrization of a point under the Gauss map. Thus, we know the distance between two infinitesimal points represented by $\vec{N}(s)$ and $\vec{N}(s')$ on Gauss circle given by

$$\|\vec{N}(s') - \vec{N}(s)\| = \|\boldsymbol{d}\vec{N}\| = \kappa(s)\,ds\,. \tag{3.24}$$

And we also have the displacement vector

$$\boldsymbol{d}\vec{r} \equiv \vec{r}(s') - \vec{r}(s) \approx \big(\cancel{\vec{r}(s)} + \vec{r}\,'(s)\,ds\big) - \cancel{\vec{r}(s)} = \vec{T}(s)\,ds \tag{3.25}$$

with the norm of

$$\|\boldsymbol{d}\vec{r}\| = \|\vec{r}(s') - \vec{r}(s)\| \approx ds\,. \tag{3.26}$$

Therefore, in the *local* region, the ratio of the length between two points on the Gauss circle and curve, i.e., $\|\boldsymbol{d}\vec{N}\|/\|\boldsymbol{d}\vec{r}\|$ can be calculate by

$$\frac{\|\vec{N}(s') - \vec{N}(s)\|}{\|\vec{r}(s') - \vec{r}(s)\|} = \frac{\|\boldsymbol{d}\vec{N}\|}{\|\boldsymbol{d}\vec{r}\|} \approx \frac{\kappa(s)\,ds}{ds} = \kappa(s)\,, \tag{3.27}$$

which measure the curvature of a curve, $\kappa(s)$, at the neighborhood of a local point $\vec{r}$.

According to the example of a circle, the curvature is $\kappa = 1/r$. So, we can assume that the position vector of the center of osculating circle is identified by the the vector

$$\vec{q} = \vec{r} + r\vec{N} = \vec{r} + (1/\kappa)\vec{N}. \tag{3.28}$$

Taking the derivative of $\vec{q}$ with respect to s, we obtain

$$\vec{q}\,' = \vec{r}\,' + \frac{1}{\kappa}\vec{N}' = \vec{T} + \frac{1}{\kappa}(-\kappa\vec{T}) = 0\,, \tag{3.29}$$

which means that $\vec{q}$ is fixed, i.e., $\vec{q}$ points to the center of the osculating circle with radius $1/\kappa$. By considering the Gauss map of a circle. The radius vector of the the Gauss circle should be $\vec{N}$, and its center C corresponds to the center of the osculating circle with position vector $\vec{q}$. Thus, in the case of a circle, the Gauss circle can be imaged as osculating circle by rescaling the length of the radius to unity.

Example (Curvature of an ellipse).

An ellipse is described by $\vec{r}(t) = \big(x(t), y(t)\big)$ with the parametrization of the coordinates $x(t) = a\cos t$ and $y(t) = b\sin t$ ($a > b > 0$), i.e.,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \cos^2 t + \sin^2 t = 1\,. \tag{3.30}$$

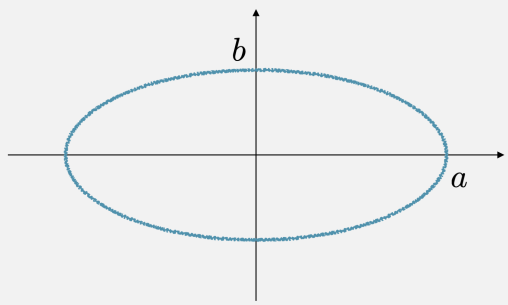

Fig. 3.4: An ellipse.

The tangent vector is

$$\dot{\vec{r}}(t) = \big(-a\sin t, b\cos t\big) \tag{3.31}$$

By changing the parameter to s, we have to calculate ds/dt first:

$$\frac{ds}{dt} = \|\dot{\vec{r}}\,\| = \sqrt{a^2\sin^2 t + b^2\cos^2 t} \quad\Longrightarrow\quad \frac{dt}{ds} = \frac{1}{\sqrt{a^2\sin^2 t + b^2\cos^2 t}} = \frac{1}{\dot{s}}\,. \tag{3.32}$$

Therefore, the tangent vector parametrized by s is obtained by

$$\begin{aligned}\vec{T} = \vec{r}\,' &= \frac{d\vec{r}}{dt}\frac{dt}{ds} \\ &= \left(\frac{-a\sin t}{\sqrt{a^2\sin^2 t + b^2\cos^2 t}}, \frac{b\cos t}{\sqrt{a^2\sin^2 t + b^2\cos^2 t}}\right) = \left(\frac{-a\sin t}{\dot{s}}, \frac{b\cos t}{\dot{s}}\right),\end{aligned} \tag{3.33}$$

Subsequently, we have

$$\vec{N} = \left(\frac{-b\cos t}{\dot{s}}, \frac{-a\sin t}{\dot{s}}\right). \tag{3.34}$$

However

$$\vec{T}\,' = \frac{d\vec{T}}{ds} = \frac{d\vec{T}}{dt}\frac{dt}{ds} = \kappa\vec{N}\,. \tag{3.35}$$

As a result, the curvature is

$$\kappa(t) = \frac{ab}{(a^2\sin^2 t + b^2\cos^2 t)^{3/2}}\,. \tag{3.36}$$

If we consider the particular case of $a = b$, an ellipse reduce to a circle with curvature $\kappa = 1/a$.

3.3 Curves in $\mathbb{R}^3$

In $\mathbb{R}^3$ we can similarly define the tangent vector $\vec{T}$ and the principal normal vector $\vec{N}$ of a point on a curve. In addition, we can define the so-called *binormal vector* through the *vector product* of $\vec{T}$ and $\vec{N}$

$$\boxed{\vec{B} = \vec{T} \times \vec{N}} \quad \text{or} \quad \boxed{\vec{B} = \vec{T} \wedge \vec{N}.}^{\dagger 4} \tag{3.37}$$

Again, we can compute the derivative of $\vec{N}$ to find the relationship between each component, i.e.,

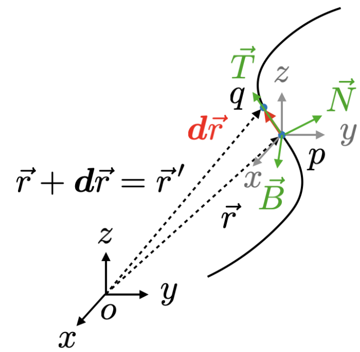

Fig. 3.5: A curve in $\mathbb{R}^3$.

the *Frenet-Serret formula* in $\mathbb{R}^3$. By differetiating the orthogonality conditions

$$\begin{cases} \vec{T} \cdot \vec{N} = 0, & (3.38\text{a}) \\ \vec{T} \cdot \vec{B} = 0, & (3.38\text{b}) \\ \vec{N} \cdot \vec{B} = 0, & (3.38\text{c}) \end{cases}$$

we obtain

$$\begin{cases} 0 = \underbrace{\vec{T}'}_{\kappa\vec{N}} \cdot \vec{N} + \vec{T} \cdot \vec{N}' = \kappa + \vec{T} \cdot \vec{N}' & (3.39\text{a}) \\ 0 = \underbrace{\vec{T}'}_{\kappa\vec{N}} \cdot \vec{B} + \vec{T} \cdot \vec{B}' = 0 + \vec{T} \cdot \vec{B}', & (3.39\text{b}) \\ 0 = \vec{N}' \cdot \vec{B} + \vec{N} \cdot \vec{B}'. & (3.39\text{c}) \end{cases}$$

Because $\{\vec{T}, \vec{N}, \vec{B}\}$ is an orthonormal frame, then we can assume that

$$\vec{N}' = a\vec{T} + b\vec{B} \quad \text{and} \quad \vec{B}' = c\vec{T} + d\vec{N}. \tag{3.40}$$

By the relations (3.39) above, we have

$$a = -\kappa, \quad c = 0 \quad \text{and} \quad d = -b. \tag{3.41}$$

Consequently $\vec{N}'$ and $\vec{B}'$ are respectively written by

$$\vec{N}' = -\kappa\vec{T} + \tau\vec{B} \quad \text{and} \quad \vec{B}' = -\tau\vec{N}, \tag{3.42}$$

[†4] The *cross product* $\times$ of two vectors cannot be used in higher dimensional space, generally we can use the *exterior product* $\wedge$ for any dimensions $n \geq 3$.

where $\tau(s) := b(s)$ is defined as the *torsion* of a curve. Then the Frenet-Serret formula is

$$\begin{cases} \vec{r}\,' = +\vec{T}, \\ \vec{T}\,' = \qquad +\kappa\vec{N}, \\ \vec{N}' = -\kappa\vec{T} \qquad +\tau\vec{B}, \\ \vec{B}\,' = \qquad -\tau\vec{N}, \end{cases} \qquad \Longrightarrow \qquad \begin{pmatrix} \vec{r}\,' \\ \vec{T}\,' \\ \vec{N}' \\ \vec{B}\,' \end{pmatrix} = \left(\begin{array}{cc:c} 1 & 0 & 0 \\ 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ \hdashline 0 & -\tau & 0 \end{array}\right) \begin{pmatrix} \vec{T} \\ \vec{N} \\ \vec{B} \end{pmatrix}. \tag{3.43}$$

Example (A helix in $\mathbb{E}^3$).

A helix is parametrized as $\vec{r} = \big(x(t), y(y), z(t)\big)$ with

$$\begin{cases} x(t) = a\cos t\,, \\ y(t) = a\sin t\,, \\ z(t) = bt\,. \end{cases} \tag{3.44}$$

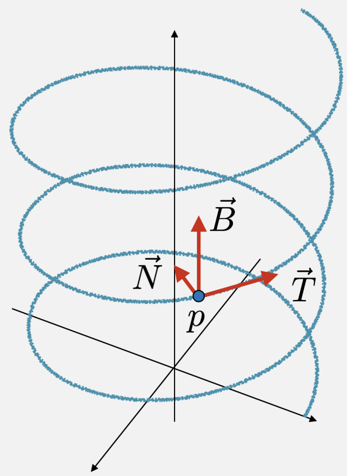

Fig. 3.6: A helix.

The tangent vector and the corresponding norm are

$$\dot{\vec{r}} = \big(\dot{x}, \dot{y}, \dot{z}\big) = \big(-a\sin t, a\cos t, b\big) \tag{3.45}$$

and

$$\|\dot{\vec{r}}\| = \sqrt{a^2\sin^2 t + a^2\cos^2 t + b^2} = \sqrt{a^2 + b^2} = \dot{s}\,. \tag{3.46}$$

The relation of s and t can be obtained by

$$s(t) = \int_0^t \frac{ds}{dt'}\,dt' = \int_0^t \sqrt{a^2+b^2}\,dt' := ct \qquad \Longrightarrow \qquad t = \frac{s}{c}\,. \tag{3.47}$$

Subsequently, we have tangent vector

$$\vec{T} = \vec{r}\,' = \left(-\frac{a}{c}\sin\frac{s}{c}, \frac{a}{c}\cos\frac{s}{c}, \frac{b}{c}\right) \tag{3.48}$$

and

$$\vec{T}' = \vec{r}'' = \left(-\frac{a}{c^2}\cos\frac{s}{c}, -\frac{a}{c^2}\sin\frac{s}{c}, 0 \right). \tag{3.49}$$

So the curvature is

$$\kappa = \|\vec{T}'\| = \frac{a}{c^2} = \frac{a}{a^2+b^2}\,. \tag{3.50}$$

and the normal vector can be obtained by

$$\vec{T}' = \kappa\vec{N} \quad\Longrightarrow\quad \vec{N} = \left(-\cos\frac{s}{c}, -\sin\frac{s}{c}, 0 \right). \tag{3.51}$$

Finally, we have binormal vector

$$\vec{B} = \vec{T}\wedge\vec{N} = \left(\frac{b}{c}\sin\frac{s}{c}, -\frac{b}{c}\cos\frac{s}{c}, \frac{a}{c} \right). \tag{3.52}$$

Due to

$$\vec{B}' = \left(\frac{b}{c^2}\cos\frac{s}{c}, \frac{b}{c^2}\sin\frac{s}{c}, 0 \right), \tag{3.53}$$

we can calculate the torsion of $\vec{r}(s)$ form (**??**):

$$\tau = \frac{b}{c^2} = \frac{b}{a^2+b^2}\,. \tag{3.54}$$

3.4 Surface Theory in $\mathbb{R}^3$

In this section we consider a 2-dimensional surface $\mathcal{M}$ realized as a set of points in $\mathbb{R}^3$, i.e., $\mathcal{M}$ is a hypersurface (codimension one) of $\mathbb{R}^3$. Any point p of a surface is represented by a position vector

$$\vec{r}(u,v) = \vec{e}_x\,x(u,v) + \vec{e}_y\,y(u,v) + \vec{e}_z\,z(u,v), \tag{3.55}$$

which parameterized by *two* variables u and v, as a vector-valued function of two variables in $\mathbb{R}^3$. Similar to the previous situation, we have p and q in $\mathbb{R}^3$, however, two points are restricted on $\mathcal{M}$ as shown in Fig. 3.7. The infinitesimal displacement vector is a *two-dimensional* vector tangent to $\mathcal{M}$ in $\mathbb{R}^3$, which can always be written as

$$\boldsymbol{d}\vec{r} = \vec{e}_x\,dx + \vec{e}_y\,dy + \vec{e}_z\,dz \equiv \boldsymbol{d}\vec{r}_{(e)}. \tag{3.56}$$

We call this expression the *local extension* of $\boldsymbol{d}\vec{r}$ to $\mathbb{R}^3$. This local extension is a higher dimensional representation particularly denoted by $\boldsymbol{d}\vec{r}_{(e)}$. Such a infinitesimal displacement vector can alternatively be written by

$$\boldsymbol{d}\vec{r} = \vec{e}_x\left(\frac{\partial x}{\partial u}du + \frac{\partial x}{\partial v}dv\right) + \vec{e}_y\left(\frac{\partial x}{\partial u}du + \frac{\partial y}{\partial v}dv\right) + \vec{e}_z\left(\frac{\partial x}{\partial u}du + \frac{\partial z}{\partial v}dv\right) \tag{3.57}$$

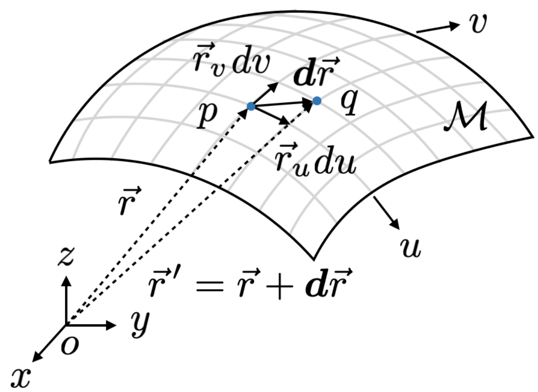

Fig. 3.7: An infinitesimal displacement vector in $\mathbb{R}^3$.

since

$$\boxed{dx^i = \frac{\partial x^i}{\partial u^a}\,du^a \quad (i,j = x,y,z \text{ and } a,b = u,v),} \tag{3.58}$$

or more precisely,

$$\begin{cases} dx = \dfrac{\partial x}{\partial u}\,du + \dfrac{\partial x}{\partial v}\,dv, & (3.59\text{a}) \\ dy = \dfrac{\partial y}{\partial u}\,du + \dfrac{\partial y}{\partial v}\,dv, & (3.59\text{b}) \\ dz = \dfrac{\partial z}{\partial u}\,du + \dfrac{\partial z}{\partial v}\,dv. & (3.59\text{c}) \end{cases}$$

The infinitesimal displacement vector through a point $p = (u, v)$ along the u-direction (with a fixed v) can be *linearly* approximated by

$$(\boldsymbol{d}\vec{r})_u := \vec{r}(u+du, v) - \vec{r}(u,v) \approx \cancel{\vec{r}(u,v)} + \frac{\partial \vec{r}(u,v)}{\partial u}\,du - \cancel{\vec{r}(u,v)} := \vec{r}_u\,du, \tag{3.60}$$

which is the infinitesimal vector with a small segment on the so-called u-*curves*. Here we have the definition

$$\vec{r}_u := \frac{\partial \vec{r}(u,v)}{\partial u} = \vec{e}_x\,\frac{\partial x}{\partial u} + \vec{e}_y\,\frac{\partial y}{\partial u} + \vec{e}_z\,\frac{\partial z}{\partial u}, \tag{3.61}$$

where we note that $\{\vec{e}_x, \vec{e}_y, \vec{e}_z\}$ is a *fixed* (or an *invariant*) frame at any point, its derivative at any point is vanished:

$$\frac{\partial \vec{e}_x}{\partial u} = \frac{\partial \vec{e}_y}{\partial u} = \frac{\partial \vec{e}_z}{\partial u} = 0. \tag{3.62}$$

Similarly, we have the displacement vector through the point p along the v-direction (with a fixed u) on the v-curves

$$(\boldsymbol{d}\vec{r})_v \approx \vec{r}_v\,dv \quad \text{with} \quad \vec{r}_v := \frac{\partial \vec{r}(u,v)}{\partial v} = \vec{e}_x\,\frac{\partial x}{\partial v} + \vec{e}_y\,\frac{\partial y}{\partial v} + \vec{e}_z\,\frac{\partial z}{\partial v}. \tag{3.63}$$

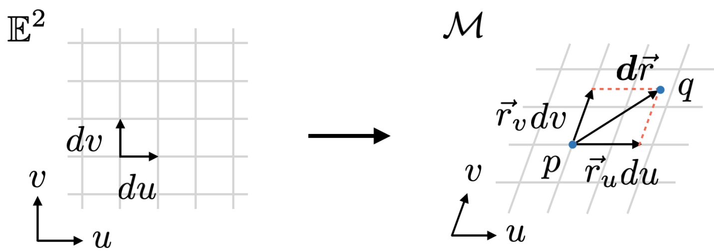

Fig. 3.8: A local coordinates (u, v) and an infinitesimal displacement vector $\boldsymbol{d}\vec{r}$ on $\mathcal{M}$.

Therefore, we obtain an approximation

$$\boxed{\boldsymbol{d}\vec{r} = (\boldsymbol{d}\vec{r})_u + (\boldsymbol{d}\vec{r})_v = \vec{r}_u\, du + \vec{r}_v\, dv = \vec{r}_a\, du^a,} \tag{3.64}$$

which is illustrated by Fig. 3.8. It is obvious that the *linearly* approximated equation above is the same as (3.56). In addition the infinitesimal displacement vector $\boldsymbol{d}\vec{r}$ is tangent at the point p, we can consequently identify the set of vector $\{\vec{r}_u, \vec{r}_v\}$ as a basis of the tangent space at p on $\mathcal{M}$.

Tangent space We call a space spanned by $\vec{r}_u$ and $\vec{r}_v$ at point p a tangent space denoted by $T_p\mathcal{M}$ with a frame $\{p; (\vec{r}_u)_p, (\vec{r}_u)_p\}$. Hence, any vector $\vec{V}_p$ can be written by

$$\vec{V}_p = (\vec{r}_u)_p \underbrace{\xi(u, v)}_{(V^u)_p} + (\vec{r}_v)_p \underbrace{\eta(u, v)}_{(V^v)_p}, \tag{3.65}$$

where the coefficients $\xi(u, v)$ and $\eta(u, v)$ are two functions of parameters u and v as the two components $(V^u)_p$ and $(V^v)_p$ of the vector $\vec{V}_p$. We note that the tangent space $T_p\mathcal{M}$ is *locally* defined at the point p. Therefore, each tangent space $T_p\mathcal{M}$ is an *internal space* at each point.

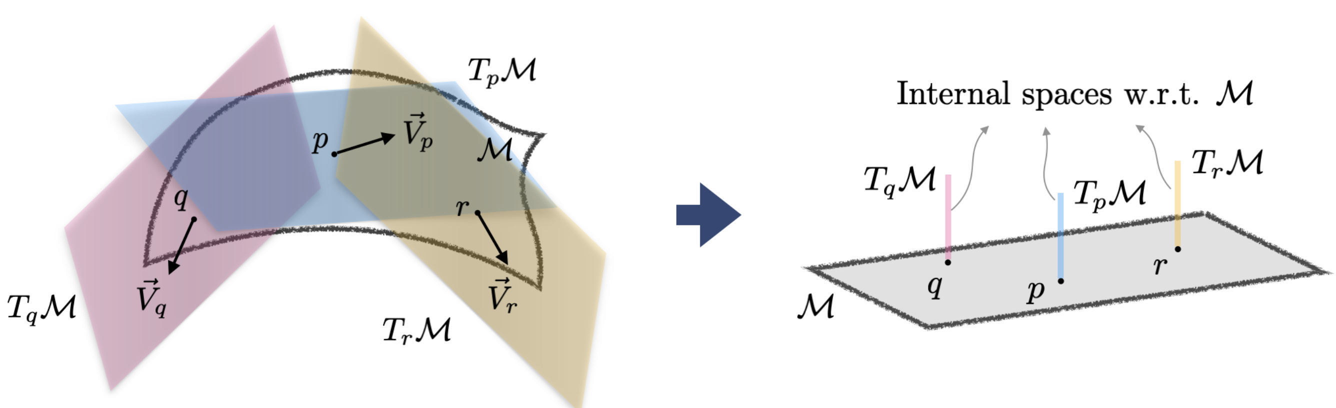

Fig. 3.9: Tangent spaces defined at three different points $p, q, r \in \mathcal{M}$.

Coordinate Transformation and Tangent Bundle

Let the tangent space to be two dimensions. A *general linear transformation* $\mathrm{GL}(2; \mathbb{R})$ (or a subgroup of $\mathrm{GL}(2; \mathbb{R})$, e.g., the *rotation* group $\mathrm{SO}(2)$ if the transformation is metric preserving) of vector $\vec{V}_p$ in the $T_p\mathcal{M}$ can be induced by a coordinate transformation such that $\widetilde{u}^a = \widetilde{u}^a(u)$. The *passive* and *active* coordinate transformation are shown in Fig. 3.10 and Fig. 3.11, respectively. However, a vector in diffferent coordinates should be equivalent, so we should have the relation:

$$\vec{V}_p = (\vec{r}_a)_p\,(V^a)_p = (\widetilde{\vec{r}}_b)_p\,(\widetilde{V}^b)_p\,. \tag{3.66a}$$

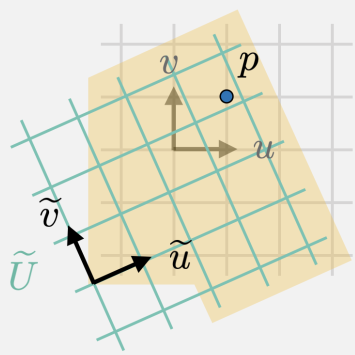

Fig. 3.10: The local coordinate trsnformation at point p.

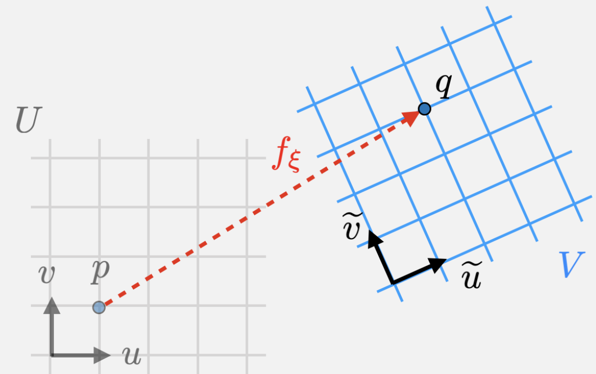

Fig. 3.11: The local translation of point p.

The coordinate transformation of the basis and components are given by

$$\begin{cases} \Big(\widetilde{V}^b(\widetilde{u})\Big)_p = \mathsf{A}^b{}_a\Big(V^a(u)\Big)_p, & (3.66b)\\ \Big(\widetilde{\vec{r}}_b(\widetilde{u})\Big)_p = \Big(\vec{r}_a(u)\Big)_p(\mathsf{A}^{-1})^a{}_b, & (3.66c)\end{cases}$$

where A and A^{-1} represent the matrices of *local coordinate transformation* (LCT) (or the *Jacobian* matrix denoted by J):

$$\mathsf{A}^b{}_a \coloneqq \frac{\partial \widetilde{u}^b}{\partial u^a} \quad \text{and} \quad (\mathsf{A}^{-1})^a{}_b = \frac{\partial u^a}{\partial \widetilde{u}^b}. \tag{3.67}$$

GCT is an internal transformation on the tangent space $T_p\mathcal{M}$ corresponding to the passive transformation in Fig. 3.10.

In general, the matrix A should be elements of the $\mathrm{GL}(2;\mathbb{R})$ transformation. The definition of (3.67) induced a particular frame transformation, so the matrix A induced by LCT is an *elements* of the $\mathrm{GL}(2;\mathbb{R})$. We should note that the definition (3.67) of the matrix A is invalid in $\mathrm{GL}(2;\mathbb{R})$. LCT does not form a group, it is just a subset of $\mathrm{GL}(2;\mathbb{R})$ rather than its subgroup.

In modern point of view, the study of the tangent vector fields on $\mathcal{M}$ can be treated as follows. We can collect all pairs of the point $p \in \mathcal{M}$ and it corresponding tangent space $T_p\mathcal{M}$. The *union* of all the tangent space $T_p\mathcal{M}$ at evry point on $\mathcal{M}$ is

$$T\mathcal{M} = \bigcup_{p\in\mathcal{M}} T_p\mathcal{M}, \tag{3.68}$$

For some kind *topological spaces*, we call the operation of the *union* of all the topological spaces the *bundle* of the topological space denoted by $\mathcal{B}$. So in current case of the union the all tangent spaces $T_p\mathcal{M}$, we have constructed a so-called *tangent bundle* $\mathcal{B} = T\mathcal{M}$. A *tangent bundle* $T\mathcal{M}$ is a special *vector bundle* denoted by a triple $\mathcal{B} = (E, B, \pi)$ with

- *base space* B: the manifold $\mathcal{M}$;
- *standard (typical) fibre* F: the space $\mathbb{R}^2$ which is *homeomorphic* to $T_p\mathcal{M}$;
- *total space* E: a collection of all $T_p\mathcal{M}$;
- *bundle projection* π (an element $\mathbf{u}$ of bundle is projected from the fibre to the corresponding base point p): $\pi(\mathbf{u}) = p$ for $\mathbf{u} \in T\mathcal{M}$;
- *structure group* G on the fibre F: $\mathrm{GL}(2;\mathbb{R})$;
- *transition function* is defined through the transformation matrix A (or J) of $\mathrm{GL}(2;\mathbb{R})$ by $g^a{}_b := \mathsf{A}^a{}_b$,

and we call $E_p = \pi^{-1}(p)$ the fibre of E over the point p (an internal space defined at p). The tangent bundle can be illustrated by Fig. 3.12.

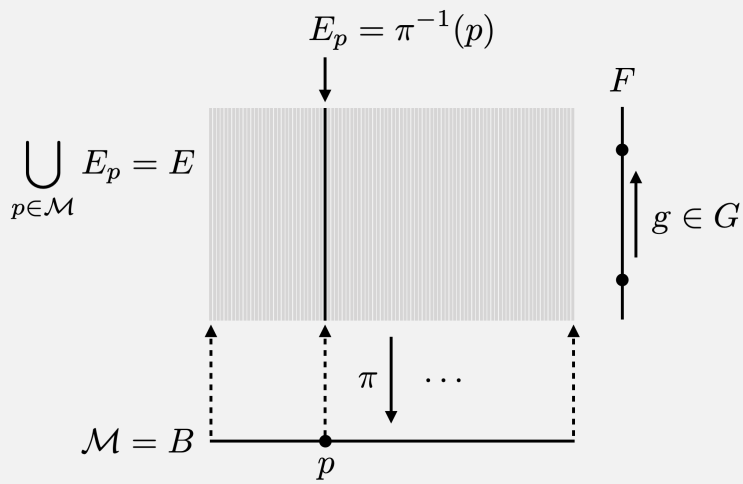

Fig. 3.12: A tangent bundle $T\mathcal{M}$ over $\mathcal{M}$.

If we take the active point of view, the generalized coordinate transformations $\widetilde{u}^a = \widetilde{u}^a(u)$ is induced from the *diffeomorphism group* $\mathrm{Diff}(\mathcal{M})$ (*differentiable homeomorphism* to $\mathcal{M}$ itself), for $p \in \mathcal{M}$ such that $q = f(p)$ with $f \in \mathrm{Diff}(\mathcal{M})$ such that u^a and $\widetilde{u}^a$ are local coordinnates of two different p and q, this is the *general corrdinate transformation* (GCT), which is the active transformation in Fig. 3.11. Then the matrix A is the group representaion $\mathrm{GL}(T\mathcal{M})$ of $\mathrm{Diff}(\mathcal{M})$ between the vector space $T_p\mathcal{M}$ and $T_{f(p)}\mathcal{M}$ in $T\mathcal{M}$ and (3.66b) is realized by

$$\left(\widetilde{V}^b(\widetilde{u})\right)_{q=f(p)} = \mathsf{A}^b{}_a(u)\left(V^a(u)\right)_p \tag{3.69}$$

It is the differential map of $\mathrm{Diff}(\mathcal{M})$ associated with an external symmetry of $\mathrm{Diff}(\mathcal{M})$ of the space $\mathcal{M}$. Furthermore, this group representaion can be extended to the group representaion on the *tensor bundle* over $\mathcal{M}$.

If we take the GCT as an *infiitesimal* transformation

$$\widetilde{u}^a = u^a + \xi^a(u), \tag{3.70}$$

it will generate a Lie algebra of the diffeomorphsim $\mathrm{Diff}(\mathcal{M})$ and the matrix $\mathsf{A} \in \mathrm{GL}(T\mathcal{M})$

$$\mathsf{A}^a{}_b = \delta^a_b + \frac{\partial \xi^a}{\partial u^b}. \tag{3.71}$$

If this infiitesimal transformation preserves the metric tensor g_{ab}, then it becomes an isometry of $\mathcal{M}$ and is reduced to SO(2), which is generated by the so-called *Killing vector* ξ^a.

A curve on a surface. Furthermore, if coordinates are functions of one parameter t or s, respectively, a curve on $\mathcal{M}$ can be obtained and parameterized either by $\gamma(t) = \vec{r}\big(x(t), y(t), z(t)\big)$ or $\gamma(s) = \vec{r}\big(u(s), v(s)\big)$. The corresponding tangent vector is respectively obtained by either

$$\dot{\vec{r}}_{(e)} = \vec{e}_x\,\dot{x} + \vec{e}_y\,\dot{y} + \vec{e}_z\,\dot{z} \quad \text{or} \quad \vec{r}\,' = \vec{r}_u\,u' + \vec{r}_v\,v'. \tag{3.72}$$

First fundamental (quadratic) form. We consider the norm of the infinitesimal displacement vector on $\mathcal{M}$ by using (3.64)

$$\begin{aligned} ds^2 &= \|\boldsymbol{d}\vec{r}\,\|^2 = \boldsymbol{d}\vec{r}\cdot\boldsymbol{d}\vec{r} \\ &= \underbrace{\vec{r}_u\cdot\vec{r}_u}_{E}\,du\,du + \underbrace{\vec{r}_u\cdot\vec{r}_v}_{F}\,du\,dv + \underbrace{\vec{r}_v\cdot\vec{r}_u}_{F}\,dv\,du + \underbrace{\vec{r}_v\cdot\vec{r}_v}_{G}\,dv\,dv \\ &= E\,du\,du + 2F\,du\,dv + G\,du\,dv \\ &= \begin{pmatrix} du & dv \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix}. \end{aligned} \tag{3.73}$$

where

$$E = \left(\frac{\partial x}{\partial u}\right)^2 + \left(\frac{\partial y}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial u}\right)^2, \tag{3.74a}$$

$$F = \frac{\partial x}{\partial u}\frac{\partial x}{\partial v} + \frac{\partial y}{\partial u}\frac{\partial y}{\partial v} + \frac{\partial z}{\partial u}\frac{\partial z}{\partial v}, \tag{3.74b}$$

$$G = \left(\frac{\partial x}{\partial v}\right)^2 + \left(\frac{\partial y}{\partial v}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2. \tag{3.74c}$$

Define dS^2 to be I, the *first fundamental* (*quadratic*) *form* of $\mathcal{M}$:

$$\boxed{\mathrm{I} := ds^2 = g_{ab}(u,v)\,du^a du^b \quad (a,b = u,v).} \tag{3.75}$$

We also note that

$$\boxed{\frac{\mathrm{I}}{ds^2} = \underbrace{g_{ab}(u,v)}_{\vec{r}_a\cdot\vec{r}_b}\,\frac{du^a}{ds}\,\frac{du^b}{ds} = \vec{r}\,'\cdot\vec{r}\,'.} \tag{3.76}$$

In addition, we have local extension $d\vec{r}_{(e)}$, hence,

$$\begin{aligned} ds^2 &= (e_i\,dx^i)\cdot(e_j\,dx^j) = g_{ij}\,dx^i dx^j \\ &= g_{ij}\left(\frac{\partial x^i}{\partial u^a}du^a\right)\left(\frac{\partial x^j}{\partial u^b}du^b\right) = \left(g_{ij}\frac{\partial x^i}{\partial u^a}\frac{\partial x^j}{\partial u^b}\right)du^a du^b. \end{aligned} \tag{3.77}$$

Then we obtain the component of the metric tensor on $\mathcal{M}$

$$\boxed{g_{ab}(u,v) \coloneqq g_{ij}(x,y,z)\frac{\partial x^i}{\partial u^a}\frac{\partial x^j}{\partial u^b} \quad (i,j = x,y,z \text{ and } a,b = u,v)} \tag{3.78}$$

called the *pullback of* $g_{ij}(x,y,z)$.

Note: In general cases, we will concern n-dimensional space $\mathcal{M}$ only and do not take $\mathcal{M}$ as a subspace of a m-dimensional Euclidean space $\mathbb{R}^m$. Therefore, there is no higher dimensional representation of the local extension of $\boldsymbol{d}\vec{r}$ to $\mathbb{R}^m$.

Orthonomal frame. By $\{\vec{r}_u, \vec{r}_v\}$, we can find a set as an orthonormal frame at p on $\mathcal{M}$ by the *Gram-Schmit* procedure. Define a operator projecting out the $\vec{e}_a$ component

$$\boxed{\vec{r}_a^{\perp} = \vec{r}_a - \sum_{b=1}^{a-1} g\big(\vec{r}_a, \vec{e}_b\big)\vec{e}_b \quad \text{and} \quad \vec{e}_b \coloneqq \frac{\vec{r}_b^{\perp}}{\|\vec{r}_b^{\perp}\|}.} \tag{3.79}$$

So the first and second basis vectors are

$$\vec{r}_u \longmapsto \vec{e}_1 \coloneqq \frac{\vec{r}_u^{\perp}}{\|\vec{r}_u^{\perp}\|} = \frac{\vec{r}_u}{\|\vec{r}_u\|} \tag{3.80}$$

and

$$\vec{r}_v \longmapsto \vec{e}_2 \coloneqq \frac{\vec{r}_v^{\perp}}{\|\vec{r}_v^{\perp}\|} = \frac{\vec{r}_v - (\vec{r}_v\cdot\vec{e}_1)\vec{e}_1}{\|\vec{r}_v - (\vec{r}_v\cdot\vec{e}_1)\vec{e}_1\|}\,, \tag{3.81}$$

Then, we can easily obtain the *unit normal vector* of $\mathcal{M}$

$$\boxed{\vec{e}_3 = \vec{e}_1 \wedge \vec{e}_2 \coloneqq \vec{n}(u,v)} \tag{3.82}$$

and its differential $d\vec{n}$ is

$$\boxed{\boldsymbol{d}\vec{n} = \frac{\partial\vec{n}}{\partial u}du + \frac{\partial\vec{n}}{\partial v}dv = \vec{n}_u\,du + \vec{n}_v\,dv = \vec{n}_a\,du^a,} \tag{3.83}$$

where we have $\vec{n}_a \coloneqq \vec{n}_{,a}$. In addition the unit normal vector $\vec{n}$ can also be obtained by

$$\boxed{\vec{n}(u,v) = \frac{\vec{r}_u \wedge \vec{r}_v}{\|\vec{r}_u \wedge \vec{r}_v\|}.} \tag{3.84}$$

Therefore, we have two sets of complete basis vector of frame $(p; \vec{r}_u, \vec{r}_u, \vec{n})$ or $(p; \vec{e}_1, \vec{e}_2, \vec{e}_3)$ in $\mathbb{R}^3$. Particularly, the frame $(p; \vec{e}_1, \vec{e}_2, \vec{e}_3)$ is an *orthonormal frame*.

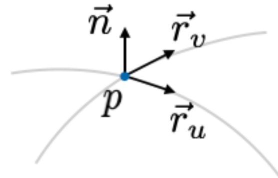

Fig. 3.13: A local frame of $\mathcal{M}$.

Second fundamental (quadratic) form. The frame $(p; \vec{r}_u, \vec{r}_u, \vec{n})$ is shown by Fig. 3.13, and we will be concentrated on such a frame for the following calculations. Vectors $\vec{r}_u$ and $\vec{r}_v$ are tangent to $\mathcal{M}$, then by relations of $\vec{n} \cdot \vec{r}_a = 0$ $(a = u, v)$, we obtain the derivative equations with respect to u^b $(b = u, v)$:

$$\underbrace{\vec{n} \cdot \vec{r}_{ab}}_{b_{ab}} + \vec{n}_b \cdot \vec{r}_a = 0 \qquad \Longrightarrow \qquad \boxed{b_{ab} \coloneqq \vec{n} \cdot \vec{r}_{ab} = -\vec{r}_a \cdot \vec{n}_b.} \tag{3.85}$$

More precisely equations are

$$\begin{cases} \vec{n} \cdot \vec{r}_{uu} = -\vec{r}_u \cdot \vec{n}_u \coloneqq L, & (3.86a) \\ \vec{n} \cdot \vec{r}_{uv} = -\vec{r}_u \cdot \vec{n}_v \coloneqq M, & (3.86b) \\ \vec{n} \cdot \vec{r}_{vu} = -\vec{r}_v \cdot \vec{n}_u \coloneqq M, & (3.86c) \\ \vec{n} \cdot \vec{r}_{vv} = -\vec{r}_v \cdot \vec{n}_v \coloneqq N. & (3.86d) \end{cases}$$

We compute a quadratic form $-\boldsymbol{d}\vec{r} \cdot \boldsymbol{d}\vec{n}$ by using (3.64), (3.83) and (3.86), then it is read as

$$\begin{aligned} -\boldsymbol{d}\vec{r} \cdot \boldsymbol{d}\vec{n} &= -(\vec{r}_u\, du + \vec{r}_v\, dv)(\vec{n}_u\, du + \vec{n}_v\, dv) \\ &= -(\vec{r}_u \cdot \vec{n}_u\, du\, du + \vec{r}_u \cdot \vec{n}_v\, du\, dv + \vec{r}_v \cdot \vec{n}_u\, dv\, du + \vec{r}_v \cdot \vec{n}_v\, dv\, dv) \\ &= \underbrace{\vec{n} \cdot \vec{r}_{uu}}_{L}\, du\, du + \underbrace{\vec{n} \cdot \vec{r}_{uv}}_{M}\, du\, dv + \underbrace{\vec{n} \cdot \vec{r}_{vu}}_{M}\, dv\, du + \underbrace{\vec{n} \cdot \vec{r}_{vv}}_{N}\, dv\, dv \\ &= L\, du\, du + 2M\, du\, dv + N\, du\, dv & (3.87a) \\ &= \begin{pmatrix} du & dv \end{pmatrix} \begin{pmatrix} L & M \\ M & N \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix} & (3.87b) \end{aligned}$$

Now the second fundamental (quadratic) form is defined by

$$\boxed{\mathbb{I} \coloneqq -\boldsymbol{d}\vec{r} \cdot \boldsymbol{d}\vec{n} = \vec{n} \cdot \boldsymbol{d}^2\vec{r} = b_{ab}\, du^a du^b.} \tag{3.88}$$

We also note that

$$\boxed{\begin{aligned} \frac{\mathbb{I}}{ds^2} &= \underbrace{b_{ab}}_{-\vec{r}_a \cdot \vec{n}_b} \frac{du^a}{ds} \frac{du^b}{ds} = -\vec{r}\,' \cdot \vec{n}\,' & (3.89) \\ &\overset{\text{or}}{=} \vec{n} \cdot \vec{r}\,''. & (3.90) \end{aligned}}$$

Gauss's formulas. When the point p move to another point q on $\mathcal{M}$, the frame also changes. The variations along u- and v-curves yield

$$\begin{cases} \boldsymbol{d}\vec{r}_u = \dfrac{\partial \vec{r}_u}{\partial u}\, du + \dfrac{\partial \vec{r}_u}{\partial v}\, dv, & (3.91a) \\ \boldsymbol{d}\vec{r}_v = \dfrac{\partial \vec{r}_v}{\partial u}\, du + \dfrac{\partial \vec{r}_v}{\partial v}\, dv. & (3.91b) \end{cases}$$

The partial derivatives of $\vec{r}_u$ and $\vec{r}_v$ with respect to u and v are new vectors, we can expand them by the frame $(p; \vec{r}_u, \vec{r}_u, \vec{n})$:

$$\begin{aligned}
\vec{r}_{uu} \coloneqq \frac{\partial \vec{r}_u}{\partial u} &= \vec{r}_u\,(\Gamma_u)^u{}_u + \vec{r}_v\,(\Gamma_u)^v{}_u + \vec{n}\,(\Gamma_u)^n{}_u \\
&= \vec{r}_u\,(\Gamma_u)^u{}_u + \vec{r}_v\,(\Gamma_u)^v{}_u + \vec{n}\,\underbrace{(\vec{n}\cdot\vec{r}_{uu})}_{L},
\end{aligned} \tag{3.92a}$$

$$\begin{aligned}
\vec{r}_{uv} \coloneqq \frac{\partial \vec{r}_u}{\partial v} &= \vec{r}_u\,(\Gamma_u)^u{}_v + \vec{r}_v\,(\Gamma_u)^v{}_v + \vec{n}\,(\Gamma_u)^n{}_v \\
&= \vec{r}_u\,(\Gamma_u)^u{}_v + \vec{r}_v\,(\Gamma_u)^v{}_v + \vec{n}\,\underbrace{(\vec{n}\cdot\vec{r}_{uv})}_{M},
\end{aligned} \tag{3.92b}$$

$$\begin{aligned}
\vec{r}_{vu} \coloneqq \frac{\partial \vec{r}_v}{\partial u} &= \vec{r}_u\,(\Gamma_v)^u{}_u + \vec{r}_v\,(\Gamma_v)^v{}_u + \vec{n}\,(\Gamma_v)^n{}_u \\
&= \vec{r}_u\,(\Gamma_v)^u{}_u + \vec{r}_v\,(\Gamma_v)^v{}_u + \vec{n}\,\underbrace{(\vec{n}\cdot\vec{r}_{vu})}_{M},
\end{aligned} \tag{3.92c}$$

$$\begin{aligned}
\vec{r}_{vv} \coloneqq \frac{\partial \vec{r}_v}{\partial v} &= \vec{r}_u\,(\Gamma_v)^u{}_v + \vec{r}_v\,(\Gamma_v)^v{}_v + \vec{n}\,(\Gamma_v)^n{}_v \\
&= \vec{r}_u\,(\Gamma_v)^u{}_v + \vec{r}_v\,(\Gamma_v)^v{}_v + \vec{n}\,\underbrace{(\vec{n}\cdot\vec{r}_{vv})}_{N}.
\end{aligned} \tag{3.92d}$$

We call these set of equations the *Gauss's formulas.* We identify the coefficients

$$\boxed{(\Gamma_a)^c{}_b \equiv \Gamma^c{}_{ab},} \tag{3.93}$$

e.g., $(\Gamma_n)^u{}_v = \Gamma^u{}_{nv}$, which is called the *affine connection coefficients* (or simply the *affinity*), and we can particularly defined a tensor called *torsion tensor* as the antisymmetric part of the affinity

$$\boxed{T^c{}_{ba} \coloneqq 2\,\Gamma^c{}_{[ab]} = \Gamma^c{}_{ab} - \Gamma^c{}_{ba}.} \tag{3.94}$$

Then Gauss formulas (3.92) now can be written as

$$\boxed{\vec{r}_{ab} = \vec{r}_c\,\Gamma^c{}_{ab} + \vec{n}\,\underbrace{\Gamma^n{}_{ab}}_{b_{ab}} \coloneqq \vec{r}_c\,\Gamma^c{}_{ab} + \vec{n}\,b_{ab} \quad (a, b, c = u, v),} \tag{3.95}$$

and the displacement vector is

$$\boxed{\boldsymbol{d}\vec{r}_a = \vec{r}_c\,\Gamma^c{}_{ab}\,du^b + \vec{n}\,\Gamma^n{}_{ab}\,du^b \coloneqq \vec{r}_c\,\Gamma^c{}_a + \vec{n}\,\Gamma^n{}_a} \tag{3.96}$$

where the coefficients are obtained by

$$\boxed{\Gamma_{cab} \coloneqq \underbrace{g_{cd}}_{\vec{r}_c\cdot\vec{r}_d}\,\Gamma^d{}_{ab} = \vec{r}_c\cdot(\underbrace{\vec{r}_d\,\Gamma^d{}_{ab}}_{\vec{r}_{ab}-\vec{n}\,b_{ab}}) = \vec{r}_c\cdot\vec{r}_{ab}} \tag{3.97}$$

and

$$\boxed{b_{ab} = \Gamma^n{}_{ab} = \vec{n}\cdot\vec{r}_{ab}.} \tag{3.98}$$

We note that Γ_{cab} and b_{ab} are *symmetric* in indices a, b.

Weingarten's formulas. Now we can also consider the partial derivative of $\vec{n}$ with respect to u and v. However, due to $\vec{n}\cdot\vec{n}=1$, we have

$$\vec{n}_a\cdot\vec{n}=0 \quad\Longrightarrow\quad \vec{n}_a\perp\vec{n}, \tag{3.99}$$

i.e., $\vec{n}_u,\vec{n}_v\perp\vec{n}$ and $\{\vec{n}_a\}$ has no component along $\vec{n}$. So we have

$$\begin{cases}\vec{n}_u=\vec{r}_u\,(\Gamma_n)^u{}_u+\vec{r}_v\,(\Gamma_n)^v{}_u\equiv\vec{r}_u\,A+\vec{r}_v\,B, & (3.100a)\\ \vec{n}_v=\vec{r}_u\,(\Gamma_n)^u{}_v+\vec{r}_v\,(\Gamma_n)^v{}_v\equiv\vec{r}_u\,C+\vec{r}_v\,D, & (3.100b)\end{cases}$$

called the *Weingarten's formulas.* We calculate the inner product of $\vec{n}_u\cdot\vec{r}_u$ and $\vec{n}_u\cdot\vec{r}_v$:

$$\begin{cases}\vec{n}_u\cdot\vec{r}_u=-L=EA+FB\\ \vec{n}_u\cdot\vec{r}_v=-M=FA+GB\end{cases}\quad\Longrightarrow\quad\boxed{A=\frac{FM-GL}{EG-F^2}}\quad\text{and}\quad\boxed{B=\frac{FL-EM}{EG-F^2}.} \tag{3.101}$$

Similarly, by computing $\vec{n}_v\cdot\vec{r}_a$, we have

$$\boxed{C=\frac{FN-GM}{EG-F^2}}\quad\text{and}\quad\boxed{D=\frac{FM-EN}{EG-F^2}.} \tag{3.102}$$

Generally the Weingarten formulas (3.100) can be written by

$$\boxed{\vec{n}_b=\vec{r}_c\,\Gamma^c{}_{nb},} \tag{3.103}$$

where the coefficients can be calculated as follows

$$\underbrace{\vec{r}_a\cdot\vec{n}_b}_{-b_{ab}}=\vec{r}_a\cdot\left(\vec{r}_c\,\Gamma^c{}_{nb}\right)=g_{ac}\,\Gamma^c{}_{nb}:=\Gamma_{anb}\quad\Longrightarrow\quad\boxed{\Gamma^c{}_{nb}=g^{ca}\Gamma_{anb}=-g^{ca}b_{ab}:=-b^c{}_b.} \tag{3.104}$$

As a result, Weingarten formulas are read as

$$\boxed{\vec{n}_b=-\vec{r}_c\,b^c{}_b.} \tag{3.105}$$

3.5 Curves on a Surface

Let variables u and v are functions of a parameter λ. then the position vector of a point on a surface becomes one dimension, which is parameterized by λ, i.e., $\vec{r}\big(u(\lambda),v(\lambda)\big)=\vec{r}(\lambda)$ becomes a curve on the surface $\mathcal{M}$. As the discussions before, it would be convenient to choose the arc length (s) as a parameter for a curve. Physically a curve $\vec{r}(s)$ represented a trajectory of an object, which is parameterized by its arc length s on $\mathcal{M}$. The corresponding velocity vector (the tangent vector) at any point on the trajectory is

$$\vec{r}\,'(s):=\frac{\boldsymbol{d}\vec{r}(s)}{ds}=\frac{\partial\vec{r}}{\partial u^a}\frac{du^a}{ds}:=\vec{r}_a\,u^{a\,\prime}, \tag{3.106}$$

and it leads to the acceleration vector is given by

$$\begin{aligned}\vec{a}=\vec{r}\,''&=\vec{r}_a'\,u^{a\,\prime}+\vec{r}_a\,u^{a\,\prime\prime}\\ &=\big(\vec{r}_c\,\Gamma^c{}_{ab}+\vec{n}\,b_{ab}\big)u^{b\,\prime}u^{a\,\prime}+\vec{r}_a u^{a\,\prime\prime}\end{aligned}$$

$$\overset{a \leftrightarrow c}{=} \underbrace{\vec{r}_a \big(u^{a\prime\prime} + \Gamma^a{}_{cb}\, u^{c\prime} u^{b\prime}\big)}_{\text{tangent}} + \underbrace{\vec{n}\, b_{ab}\, u^{a\prime} u^{b\prime}}_{\text{normal}} \tag{3.107a}$$

$$\equiv \vec{a}_t + \vec{a}_n \tag{3.107b}$$

$$:= \vec{\kappa}_g + \vec{\kappa}_n, \tag{3.107c}$$

where we have used the chain rule and the Gauss's formulas to have

$$\vec{r}_a^{\,\prime} = \vec{r}_{ab} \frac{du^b}{ds} = \big(\vec{r}_c\, \Gamma^c{}_{ab} + \vec{n}\, b_{ab}\big) u^{b\prime}. \tag{3.108}$$

Geometrically the acceleration defines the curvature of a curve, therefore, the tangent and normal acceleration vectors $\vec{a}_t$ and $\vec{a}_n$ are corresponding to the so-called *geodesic* and *normal curvature vectors* $\vec{\kappa}_g$ and $\vec{\kappa}_n$, respectively. The magnitudes of two curvature vectors $\kappa_g := \|\vec{\kappa}_g\| = \|\vec{a}_t\|$ and $\kappa_n := \|\vec{\kappa}_n\| = \|\vec{a}_n\|$ are called the *geodesic curvature* and *normal curvature* respectively.

Geodesic equations. Now we call a curve $\vec{r}(s)$ *geodesic* if the curve is a *straight line* and its acceleration is just the normal curvature vector $\vec{r}^{\,\prime\prime} = \vec{a}_n =: \vec{\kappa}_n$, i.e., the *tangent* part of the acceleration vector *vanishes*

$$\boxed{\vec{a}_t = \vec{r}_a \big(u^{a\prime\prime} + \Gamma^a{}_{cb}\, u^{c\prime} u^{b\prime}\big) = 0.} \tag{3.109}$$

The geodesic can also be realized as follows. A curve $\vec{r}(s)$ *only* has the normal curvature $\vec{\kappa}_n$, which *only* changes the direction of a moving object. For instance, the trajectory of a satellite is circular. The centripetal force is used to change the direction of the satellite only.

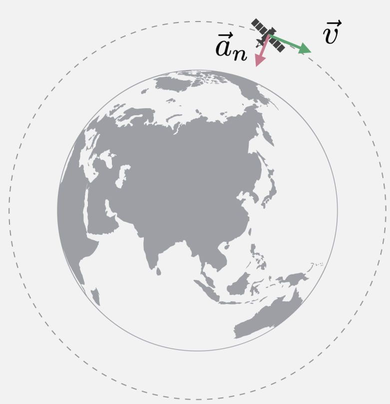

Fig. 3.14: The circular motion of a satellite.

That means the *tangent* velocity $\vec{V}_t = (\vec{r}^{\,\prime})_t$ of the object is *constant* and the motion on $\mathcal{M}$ satisfies the law of inertia of Galileo. Thus the object only feels a force *vertically*. Because $\vec{r}_a$s are linearly independent, (3.109) is reduced to

$$\boxed{u^{a\prime\prime} + \Gamma^a{}_{cb}\, u^{c\prime} u^{b\prime} = 0,} \tag{3.110}$$

which is called the *geodesic equation*.

Normal curvature. It is obvious that, from (3.88), we can compute the ratio of the second and the first fundamental form of a surface:

$$\frac{\mathrm{I\!I}}{\mathrm{I}} = \frac{\mathrm{I\!I}}{ds^2} = \vec{n}\cdot\frac{\boldsymbol{d}^2\vec{r}}{ds^2} = \vec{n}\cdot\vec{a} = a_n \overset{\text{or}}{=} \kappa_n \tag{3.111}$$

is the *normal* curvature. We assume that κ_n has value of λ, which leads to the relation

$$\mathrm{I\!I} = \lambda\,\mathrm{I}. \tag{3.112}$$

One can divide (3.112) by ds^2 and then, with the help of definitions (3.75) and (3.88), the equation becomes a differential equation:

$$\frac{\mathrm{I\!I}}{ds^2} = \lambda\,\frac{\mathrm{I}}{ds^2} \quad\Longrightarrow\quad L\,u'u' + 2M\,u'v' + N\,v'v' = \lambda\big(Eu'u' + 2Fu'v' + Gv'v'\big), \tag{3.113}$$

where λ can be recognized as the Lagrangian multiplier with the constraint

$$Eu'u' + 2Fu'v' + Gv'v' = 1 \quad \text{due to} \quad \frac{\mathrm{I}}{ds^2} = 1. \tag{3.114}$$

We note that the left hand side of (3.113) is κ_n due to (3.111). Therefore, solving the equation of (3.113) is to find the values of λ to extremize κ_n, we take the partial derivative of (3.113) with respect to $u^{i\prime}$, which leads to the equation of matrix form

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix}\begin{pmatrix} u' \\ v' \end{pmatrix} = \lambda\begin{pmatrix} E & F \\ F & G \end{pmatrix}\begin{pmatrix} u' \\ v' \end{pmatrix} \quad\Longrightarrow\quad \begin{pmatrix} L-\lambda E & M-\lambda F \\ M-\lambda F & N-\lambda G \end{pmatrix}\begin{pmatrix} u' \\ v' \end{pmatrix} = 0\,, \tag{3.115}$$

which means

$$\boxed{(b_{ab} - \lambda g_{ab})u^{b\prime} = 0\,.} \tag{3.116}$$

We have to look for the *non-trivial* solutions, i.e.,

$$\begin{aligned} & \det(b_{ab} - \lambda g_{ab}) = 0\,, \\ \Longrightarrow\quad & \big(EG - F^2\big)\lambda^2 - \big(EN + GL - 2FM\big)\lambda + LN - M^2 = 0\,, \\ \Longrightarrow\quad & \mathbf{g}\lambda^2 - \big(EN + GL - 2FM\big)\lambda + \mathbf{b} = 0\,, \end{aligned} \tag{3.117}$$

where we define

$$\begin{cases} \mathbf{g} := \det(g_{ab}) = EG - F^2\,, & \text{(3.118a)} \\ \mathbf{b} := \det(b_{ab}) = LN - M^2\,. & \text{(3.118b)} \end{cases}$$

As a result, we have the sum and product of two solutions λ_1 and λ_2

$$\lambda_1 + \lambda_2 = \frac{EN + GL - 2FM}{\mathbf{g}} \quad \text{and} \quad \lambda_1\lambda_2 = \frac{\mathbf{b}}{\mathbf{g}}\,. \tag{3.119}$$

The value of λ_α $(\alpha = 1, 2)$ is called the *principal curvature* of κ_n. By substituting λ_α into the equation (3.116), the corresponding solution of the tangent vector $\vec{r}\,'_{(\alpha)} = \vec{r}_a\,u^{a\prime}_{(\alpha)}$ or $\boldsymbol{d}\vec{r}_{(\alpha)} = \vec{r}_a\,du^a_{(\alpha)}$ is called the *principal direction*.

Gauss curvature. We define the *Gauss curvature* (or called *total curvature*) as the product of two curvatures:

$$\boxed{K := \lambda_1 \lambda_2 = \frac{\mathbf{b}}{\mathbf{g}}\,.} \tag{3.120}$$

Mean curvature. The *mean curvature* is defined by the mean value of the sum of two curvatures:

$$\boxed{H := \frac{1}{2}(\lambda_1 + \lambda_2) = \frac{EN + GL - 2FM}{2\,\mathbf{g}}\,.} \tag{3.121}$$

Example (A cylindrical surface in $\mathbb{E}^3$).

A surface parallel with z-axis can be described by

$$\vec{r} = \big(x, y, z\big) = \big(x(u), y(u), v\big) \quad \Longrightarrow \quad \boldsymbol{d}\vec{r} = \big(dx, dy, dz\big) = \big(x'du, y'du, dv\big)\,. \tag{3.122}$$

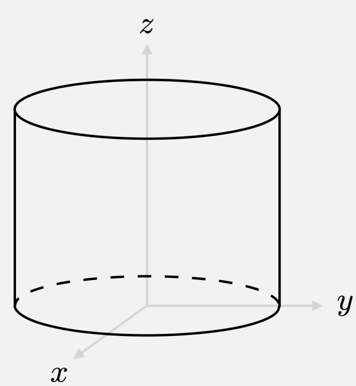

Fig. 3.15: A cylinder.

A cylindrical surface need to have a constraint with

$$\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2 = x'^2 + y'^2 = 1\,. \tag{3.123}$$

A cylinder is parametrized by

$$\vec{r}(u, v) = \big(x(u), y(u), v\big) \quad \Longrightarrow \quad \boldsymbol{d}\vec{r} = \vec{r}_u du + \vec{r}_v dv = \big(x'du, y'du, dv\big)\,, \tag{3.124}$$

where

$$\begin{cases} \vec{r}_u = \big(x', y', 0\big)\,, \\ \vec{r}_v = \big(0, 0, 1\big)\,. \end{cases} \quad \Longrightarrow \quad \vec{r}_u \wedge \vec{r}_v = \big(y', -x', 0\big)\,. \tag{3.125}$$

and the normal vector is

$$\vec{n} = \frac{\vec{r}_u \wedge \vec{r}_v}{|\vec{r}_u \wedge \vec{r}_v|} = \frac{1}{y'^2 + x'^2}\big(y', -x', 0\big) = \big(y', -x', 0\big) \tag{3.126}$$

and

$$d\vec{n} = \vec{n}_u du + \vec{n}_v dv = \big(y'' du, -x'' du, 0\big)\,. \tag{3.127}$$

with

$$\begin{cases} \vec{n}_u = \big(y'', -x'', 0\big)\,, \\ \vec{n}_v = \big(0, 0, 0\big)\,. \end{cases} \tag{3.128}$$

Then we have first fundamental form

$$\mathrm{I} = \boldsymbol{d}\vec{r} \cdot \boldsymbol{d}\vec{r} = \big(x'^2 + y'^2\big) du^2 + dv^2 = du^2 + dv^2\,, \tag{3.129}$$

where

$$E = G = 1, \quad F = 0\,. \tag{3.130}$$

The second fundamental form can be obtained by

$$\begin{aligned} \mathrm{II} &= -\boldsymbol{d}\vec{r} \cdot d\vec{n} \\ &= -(\vec{r}_u \cdot \vec{n}_u\, dudu + \underbrace{\vec{r}_u \cdot \vec{n}_v}_{0}\, dudv + \underbrace{\vec{r}_v \cdot \vec{n}_u}_{0}\, dudv + \underbrace{\vec{r}_v \cdot \vec{n}_v}_{0}\, dvdv) \\ &= -\big(x'y'' - y'x''\big) du^2 \\ &= \big(y'x'' - x'y''\big) du^2\,, \end{aligned} \tag{3.131}$$

where

$$L = y'x'' - x'y'', \quad M = N = 0\,. \tag{3.132}$$

So we have

$$\mathbf{g} = 1\,, \quad \mathbf{b} = 0\,, \quad EN + GL - 2FM = y'x'' - x'y''\,. \tag{3.133}$$

As a result, we obtain

$$\text{Gauss curvature:} \quad K = \lambda_1 \lambda_2 = 0, \tag{3.134}$$

$$\text{mean curvature:} \quad H = \frac{1}{2}\big(\lambda_1 + \lambda_2\big) = \frac{1}{2}\big(y'x'' - x'y''\big)\,. \tag{3.135}$$

We can solve the above equations to obtain

$$\lambda_1 = 0, \quad \text{and} \quad \lambda_2 = y'x'' - x'y''\,. \tag{3.136}$$

3.6 Covariant Derivative and Parallel Transport

Covariant derivative. Generally, given an arbitrary vector $\vec{X} = \vec{r}_a\, X^a$, we have

$$\frac{\boldsymbol{d}\vec{X}}{ds} = \vec{X}' = \vec{r}_a^{\,\prime}\, X^a + \vec{r}_a\, X^{a\,\prime}$$

$$= \underbrace{\vec{r}_a\big(X^{a\prime} + \Gamma^a{}_{cb}\, X^c u^{b\prime}\big)}_{\text{tangent}} + \underbrace{\vec{n}\, b_{ab}\, X^a u^{b\prime}}_{\text{normal}}. \tag{3.137}$$

We can use the chain rule to rewrite $X^{a\prime}$ by

$$X^{a\prime} = \frac{\partial X^a}{\partial u^b}\frac{du^b}{ds} = X^a{}_{,b}\, u^{b\prime} \tag{3.138}$$

as a product of a partial derivative with respect to the coordinates u^b and $u^{b\prime}$ by chain rule. Substituting the expression into (3.137), then $\vec{X}'$ would be read by

$$\vec{X}' = \underbrace{\vec{r}_a\big(X^a{}_{,b} + \Gamma^a{}_{cb}\, X^c\big)u^{b\prime}}_{(\vec{X}')_t := (\vec{X}_b)_t\, u^{b\prime}} + \underbrace{\vec{n}\, b_{ab}\, X^a u^{b\prime}}_{(\vec{X}')_n := (\vec{X}_b)_n\, u^{b\prime}} := \Big[\big(\vec{X}_b\big)_t + \big(\vec{X}_b\big)_n\Big]u^{b\prime}, \tag{3.139}$$

where t and n denote the *tangent* and *normal* parts respectively. We define the *tangent* part of the derivative of $\vec{X}$ by a new differentiation $\boldsymbol{D}$:

$$\boxed{\big(\vec{X}'\big)_t := \frac{\boldsymbol{D}\vec{X}}{ds} := \boldsymbol{D}_{\vec{r}\,\prime}\vec{X} := u^{b\prime}\big(\boldsymbol{D}_b\vec{X}\big),} \tag{3.140a}$$

where

$$\boxed{\frac{\boldsymbol{D}}{ds} := \boldsymbol{D}_{\vec{r}\,\prime} = \boldsymbol{D}_{(\vec{r}_b u^{b\prime})} = u^{b\prime}\boldsymbol{D}_{\vec{r}_b} \equiv u^{b\prime}\boldsymbol{D}_b\,,} \tag{3.140b}$$

which is the directional derivative along the direction of the tangent vector $\vec{r}\,' = \vec{r}_b\, u^{b\prime}$ of a trajectory $u^b(s)$ on $\mathcal{M}$ and

$$\boldsymbol{D}_b\vec{X} \equiv \vec{r}_a\big(\boldsymbol{D}_b\vec{X}\big)^a := \big(\vec{X}_b\big)_t = \vec{r}_a\big(X^a{}_{,b} + \Gamma^a{}_{cb}\, X^c\big), \tag{3.141}$$

is the so-called *covariant derivative* of $\vec{X}$ with respect to the coordinate u^b as the *tangent part* of $\vec{X}_{,b}$ on $\mathcal{M}$ by using the *bold* symbol $\boldsymbol{D}$. We note that the action of the *covariant derivative operator* $\boldsymbol{D}_b$ on the vector $\vec{X}$ is

$$\boldsymbol{D}_b\vec{X} = \boldsymbol{D}_b\big(\vec{r}_a\, X^a\big) = \big(\boldsymbol{D}_b\vec{r}_a\big)X^a + \vec{r}_a\big(\boldsymbol{D}_b X^a\big). \tag{3.142}$$

Hence, by comparing (3.141) and (3.142), the action of $\boldsymbol{D}_b$ on basis of $\vec{r}_a$ and component X^a respectively are

$$\boxed{\begin{cases} \boldsymbol{D}_b\vec{r}_a = (\vec{r}_{ab})_t = \vec{r}_c\,\Gamma^c{}_{ab}\,, & (3.143\text{a}) \\ \boldsymbol{D}_b X^a = \partial_b X^a \overset{\text{or}}{=} X^a{}_{,b}\,. & (3.143\text{b}) \end{cases}}$$

Alternatively we can define the covariant derivative in another form denoted by ∇ with different differential properties, which is always used by physicists to work on the component computations in the *tensor calculus*:

$$\boxed{\begin{cases} \nabla_b X^a \equiv X^a{}_{;b} := \big(\boldsymbol{D}_b\vec{X}\big)^a = X^a{}_{,b} + \Gamma^a{}_{cb}\, X^c, & (3.144\text{a}) \\ \nabla_b\vec{r}_a \equiv \vec{r}_{a;b} = 0. & (3.144\text{b}) \end{cases}}$$

As a result, $\nabla_b X^a$ is the *component expression* of $\boldsymbol{D}_b\vec{X}$ in the direction of $\vec{r}_a$. Therefore, we note that ∇ has nothing to do on the basis vectors. The equivalence of two operators can be clarified as follows:

$$\boldsymbol{D}_b\vec{X} = \vec{r}_a(\boldsymbol{D}_b\vec{X})^a = \vec{r}_a(\underbrace{\nabla_b X^a}_{X^a{}_{;b}}) + \overbrace{0}^{0\cdot X^a} \underset{\uparrow\ \boxed{0 = \nabla_b \vec{r}_a \overset{\text{or}}{=} \vec{r}_{a;b}}}{:=} \nabla_b\vec{X} \overset{\text{or}}{=} \vec{X}_{;b} \tag{3.145}$$

such that

$$\boxed{\frac{\nabla\vec{X}}{ds} = \vec{r}_a\frac{\nabla X^a}{ds} := \vec{r}_a(\nabla_{\vec{r}\,'}X^a) = \vec{r}_a(u^{b\prime}\,\nabla_b X^a) \overset{\text{or}}{=} \vec{r}_a\big(X^a{}_{;b}\,u^{b\prime}\big).} \tag{3.146}$$

Parallel transport. We will say that any vector $\vec{X}$ is parallel to the curve $\vec{r}(s)$ if its *tangent* part of the derivative vanishes

$$\big(\vec{X}'\big)_t = \frac{\boldsymbol{D}\vec{X}}{ds} = \frac{\nabla\vec{X}}{ds} = \vec{r}_a\frac{\nabla X^a}{ds} = \vec{r}_a\big(u^{b\prime}\,\nabla_b X^a\big) = 0 \quad\Longrightarrow\quad \boxed{u^{b\prime}\,\nabla_b X^a = 0.} \tag{3.147}$$

Since $u^{b\prime}$ is arbitrary but non-zero, the so-called *parallel transport* of a vector X^a can also be written as

$$\boxed{\nabla_b X^a = \partial_b X^a + \Gamma^a{}_{cb}X^c = 0} \quad \text{with the operator} \quad \boxed{\big(\nabla_b\big)^a{}_c = \delta^a_c\partial_b + \Gamma^a{}_{cb}} \tag{3.148a}$$

or

$$\boxed{\partial_b X^a = -\Gamma^a{}_{cb}X^c.} \tag{3.148b}$$

Parallel transport of a vector can be realized as follows. Let a curve $\vec{r}(s)$ parameterized by s. If

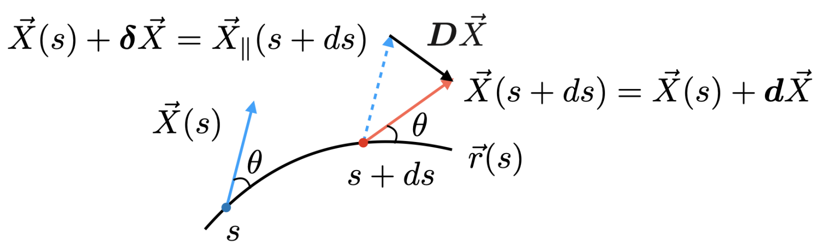

Fig. 3.16: Parallel transport of vector $\vec{X}$.

we are interested in a vector $\vec{X}(s)$ parallel transported from s to $s+ds$, we have to look for the relation between vectors $\vec{X}(s)$ and $\vec{X}(s+ds)$. Let the transported vector at $s+ds$ defined by

$$\vec{X}_{\parallel}(s+ds) = \vec{X}(s) + \boldsymbol{\delta}\vec{X}, \tag{3.149}$$

where $\boldsymbol{\delta}\vec{X}$ should be linear in the change of coordinate and the vector itself, then it is given by

$$\boxed{\boldsymbol{\delta}\vec{X} = -\vec{r}_a\,\underbrace{\Gamma^a{}_{cb}\,X^c\,du^b}_{\delta X^c}.} \tag{3.150}$$

By the way, the vector defined at $s+ds$ can be expanded as

$$\vec{X}(s+ds) = \vec{X}(s) + \boldsymbol{d}\vec{X} \quad \text{with} \quad \boxed{\boldsymbol{d}\vec{X} = \vec{X}'\,ds = \vec{r}_a\,\underbrace{X^a{}_{,b}\,du^b}_{dX^a}.} \tag{3.151}$$

Then we define $\boldsymbol{D}\vec{X}$ by

$$\boxed{\begin{aligned}\boldsymbol{D}\vec{X} &:= \vec{X}(s+ds) - \vec{X}_{\parallel}(s+ds)\\ &= \boldsymbol{d}\vec{X} - \boldsymbol{\delta}\vec{X},\end{aligned}} \tag{3.152}$$

which is illustrated in Fig. 3.16. If $\vec{X}(s)$ is parallel transported from s to $s+ds$, it means that the angle θ between $\vec{X}$ and $\vec{r}(s)$ is preserved and the transported vector should be equal to the vector defined at the transported point at $s+ds$, i.e.,

$$\boxed{\vec{X}(s+ds) = \vec{X}_{\parallel}(s+ds)} \quad \Longrightarrow \quad \boxed{\boldsymbol{D}\vec{X} = 0} \quad \text{or} \quad \boxed{\boldsymbol{d}\vec{X} = \boldsymbol{\delta}\vec{X}.} \tag{3.153}$$

So we deserves the same equation as (3.148a). Particularly, we call a vector $\vec{X}$ *uniform* if $\vec{X}$ satisfies the condition of the parallel transport $\boldsymbol{D}\vec{X} = 0$.

Furthermore, if a vector $\vec{X} = \vec{r}\,'(s)\vec{r}_a\,u^{a\,\prime} := \vec{r}_a V^a = \vec{V}$ is the tangent vector of a geodesic, the component of the equation of the parallel transport of a vector $\vec{X} = \vec{V}$ is

$$\left(\vec{V}'\right)^a_t = \left(\frac{\boldsymbol{D}\vec{V}}{ds}\right)^a = \frac{\nabla V^a}{ds} = V^b\left(\nabla_b V^a\right) = 0, \tag{3.154}$$

and by the definition of the covariant derivative, we found

$$u^{a\,\prime\prime} + \Gamma^a{}_{cb}\,u^{c\,\prime}u^{b\,\prime} = u^{b\,\prime}\frac{\partial u^{a\,\prime}}{\partial u^b} + \Gamma^a{}_{cb}\,u^{c\,\prime}u^{b\,\prime} = V^a{}_{,b}\,V^b + \Gamma^a{}_{cb}\,V^c V^b = V^b\nabla_b V^a, \tag{3.155}$$

thus the geodesic equations (3.110) can be rewritten as

$$\boxed{V^b\nabla_b V^a \equiv V^a{}_{;b}\,V^b = 0,} \tag{3.156}$$

which coincides with the equation of parallel transport.

Note: As we have mentioned that we will always concern a n-dimensional space $\mathcal{M}$ which is not a subspace of other space, so we have no second fundamental form for $\mathcal{M}$, i.e., $b_{ij} = 0$, and $\vec{V}' = \left(\vec{V}'\right)_t$ is tangent to $\mathcal{M}$, indeed. Hence, the derivative of a tangent vector should be

$$\begin{aligned}\vec{V}' &= \frac{\boldsymbol{D}\vec{V}}{ds} = V^i\,\boldsymbol{D}_i\vec{V}\\ &\overset{\text{or}}{=} e_j\,\frac{\nabla V^j}{ds} = e_j\left(V^i\,\nabla_i V^j\right) \quad (i,j = 1,2,\cdots,n).\end{aligned} \tag{3.157}$$

Covariant derivative of a metric tensor. We assume that the magnitude of $\vec{X}$ is preserved under the parallel transport, additionally the norm $\|\vec{X}\|^2$ is a scalar, we would have

$$
\begin{aligned}
0 &= \frac{d}{ds}\|\vec{X}\|^2 = \frac{\nabla}{ds}\|\vec{X}\|^2 \\
&= \left(u^{c\prime}\,\nabla_c\right)\left(g_{ab}X^aX^b\right) \\
&= \left(u^{c\prime}\,\nabla_c\, g_{ab}\right)X^aX^b + g_{ab}\big(\underbrace{u^{c\prime}\,\nabla_c X^a}_{0}\big)X^b + g_{ab}X^a\big(\underbrace{u^{c\prime}\,\nabla_c X^b}_{0}\big).
\end{aligned}
\tag{3.158}
$$

The result gives the covariant derivative of a metric tensor

$$
\boxed{\nabla_c\, g_{ab} = 0,}
\tag{3.159}
$$

which is the well-known *metric compatible condition* or *metricity condition*. It implies that the metric of a space is preserved under the parallel transport. It can be shown that the *cosine* between $\vec{X} = \vec{r}_a X^a$ and tangent vector $\vec{r}\,' = \vec{r}_b\, u^{b\prime}$, which stands for by $\vec{r}_a \cdot \vec{r}_b = \|\vec{r}_a\|\|\vec{r}_b\|\cos\theta = g_{ab}$, stays the *same* under the parallel transport:

$$
\frac{\nabla}{ds}\Big(\vec{r}_a\cdot\vec{r}_b\Big) = \frac{\nabla}{ds}\Big(\|\vec{r}_a\|\|\vec{r}_b\|\cos\theta\Big) = \underbrace{\frac{d\|\vec{r}_a\|}{ds}}_{0}\,\|\vec{r}_b\|\,\cos\theta + \|\vec{r}_a\|\,\underbrace{\frac{d\|\vec{r}_b\|}{ds}}_{0}\,\cos\theta + \|\vec{r}_a\|\|\vec{r}_a\|\,\frac{d\cos\theta}{ds}
\tag{3.160}
$$

However,

$$
\frac{\nabla}{ds}\Big(\vec{r}_a\cdot\vec{r}_b\Big) = \left(u^{c\prime}\,\nabla_c\right) g_{ab} = u^{c\prime}\,\underbrace{\nabla_c\, g_{ab}}_{0} = 0.
\tag{3.161}
$$

So, we would have

$$
\frac{d\cos\theta}{ds} = 0 \quad\Longrightarrow\quad \boxed{\theta = \text{const.}}
\tag{3.162}
$$

This consequence can be illustrated by Fig. 3.16 through $\boldsymbol{D}\vec{X} = 0$.

On the other hand, we can do the derivative of $\|\vec{X}\|^2$ with respect to s directly:

$$
\begin{aligned}
\frac{d}{ds}\|\vec{X}\|^2 &= \frac{d}{ds}\left(g_{ab}X^aX^b\right) \\
&= u^{c\prime}\left(\partial_c\, g_{ab} - g_{db}\Gamma^d{}_{ac} - g_{ad}\Gamma^d{}_{bc}\right)X^aX^b.
\end{aligned}
\tag{3.163}
$$

As a result, we can equal two results and obtain the rule for calculating the covariant derivative of a metric tensor

$$
\boxed{
\begin{aligned}
\nabla_c\, g_{ab} &= \partial_c\, g_{ab} - \Gamma^d{}_{ac}\, g_{db} - \Gamma^d{}_{bc}\, g_{ad} \\
&= \partial_c\, g_{ab} - \Gamma_{bac} - \Gamma_{abc}.
\end{aligned}
}
\tag{3.164}
$$

It can be shown that the covariant derivative of g^{ab} should be

$$
\boxed{\nabla_c\, g^{ab} = \partial_c\, g^{ab} + \Gamma^a{}_{dc}\, g^{db} + \Gamma^b{}_{dc}\, g^{ad}.}
\tag{3.165}
$$

Covariant derivative of an arbitrary rank tensor. We can construct of a set of basis vector of $\{du^a\}$ in $T_p^*\mathcal{M}$, where $T_p^*\mathcal{M}$ is the dual space of $T_p\mathcal{M}$ and $\{du^a\}$ is called the dual basis vector of $\{\vec{r}_b\}$such that

$$\boxed{du^a(\vec{r}_b) = \delta_b^a} \quad \text{and} \quad \boxed{\boldsymbol{D}_c\, du^a = -\Gamma^a{}_{bc}\, du^b.} \tag{3.166}$$

We define an arbitrary rank tensor, for example a (r, s)-type tensor by

$$K^{a_1 a_2 \cdots a_r}{}_{b_1 b_2 \cdots b_s}\, \vec{r}_{a_1} \otimes \vec{r}_{a_2} \otimes \cdots \otimes \vec{r}_{a_r} \otimes du^{b_1} \otimes du^{b_2} \otimes \cdots \otimes du^{b_s}. \tag{3.167}$$

Particularly, two tensors

$$\begin{cases} (1,0)\text{-type:} \quad K = K^{a_1}\, \vec{r}_{a_1}, \\ (0,1)\text{-type:} \quad K = K_{b_1}\, du^{b_1} \end{cases} \tag{3.168}$$

are the vector and covector respectively.

its covariant derivative can be generalized from the result of the metric tensor:

$$\boxed{\begin{aligned} \nabla_c K^{a_1 a_2 \cdots a_r}{}_{b_1 b_2 \cdots b_s} = \partial_c K^{a_1 a_2 \cdots a_r}{}_{b_1 b_2 \cdots b_s} & \\ + \Gamma^{a_1}{}_{dc}\, K^{d a_2 \cdots a_r}{}_{b_1 b_2 \cdots b_s} + \Gamma^{a_2}{}_{dc}\, K^{a_1 d \cdots a_r}{}_{b_1 b_2 \cdots b_s} & \\ + \cdots + \Gamma^{a_r}{}_{dc}\, K^{a_1 a_2 \cdots d}{}_{b_1 b_2 \cdots b_s} & \\ - \Gamma^{d}{}_{b_1 c}\, K^{a_1 a_2 \cdots a_r}{}_{d b_2 \cdots b_s} - \Gamma^{d}{}_{b_2 c}\, K^{a_1 a_2 \cdots a_r}{}_{b_1 d \cdots b_s} & \\ - \cdots - \Gamma^{d}{}_{b_s c}\, K^{a_1 a_2 \cdots a_r}{}_{b_1 b_2 \cdots d}. & \end{aligned}} \tag{3.169}$$

<u>**Note:**</u> It is easy to obtain the result of $\nabla_c K^{a_1 a_2 \cdots a_r}{}_{b_1 b_2 \cdots b_s}$ by applying $\boldsymbol{D}$ to the tensor K:

$$\boxed{\boldsymbol{D}_c K = \left(\nabla_c K^{a_1 a_2 \cdots a_r}{}_{b_1 b_2 \cdots b_s}\right) \vec{r}_{a_1} \otimes \vec{r}_{a_2} \otimes \cdots \otimes \vec{r}_{a_r} \otimes du^{b_1} \otimes du^{b_2} \otimes \cdots \otimes du^{b_s}.} \tag{3.170}$$

3.7 Christoffel Symbol and Curvature Tensor

According to (3.97), we can obtain the following equations

$$\begin{cases} \dfrac{\partial}{\partial u^c}\, g_{ab} = \dfrac{\partial}{\partial u^c}(\vec{r}_a \cdot \vec{r}_b) = \vec{r}_{ac} \cdot \vec{r}_b + \vec{r}_a \cdot \vec{r}_{bc} = \Gamma_{bac} + \Gamma_{abc}\,, & \text{(3.171a)} \\ \dfrac{\partial}{\partial u^a}\, g_{bc} = \Gamma_{cba} + \Gamma_{bca}\,, & \text{(3.171b)} \\ \dfrac{\partial}{\partial u^b}\, g_{ca} = \Gamma_{acb} + \Gamma_{cab}\,. & \text{(3.171c)} \end{cases}$$

Equations above are, in fact, the metricity condition of the covariant derivative of g_{ab}. Through the combination (3.171c) + (3.171b) − (3.171a), with the help of symmetry

$$\vec{r}_{ab} = \frac{\partial^2 \vec{r}}{\partial u^a \partial u^b} = \vec{r}_{ba}, \tag{3.172}$$

the specific solution of the affinity Γ_{cab} is a function of the metric tensor g_{ab}:

$$\boxed{\Gamma_{cab} = [c, ab],} \tag{3.173}$$

where the shorthand notation

$$\boxed{[c, ab] \coloneqq \frac{1}{2}\left(\frac{\partial}{\partial u^a}\, g_{bc} + \frac{\partial}{\partial u^b}\, g_{ca} - \frac{\partial}{\partial u^c}\, g_{ab}\right)} \tag{3.174}$$

is called the *Christoffel symbol of the first kind.*[†5] For the first-index raised affinity $\Gamma^d{}_{ab} = g^{dc}\Gamma_{cab}$, we have

$$\boxed{\Gamma^d{}_{ab} = g^{dc}[c, ab] \coloneqq \left\{ {d \atop ab} \right\},} \tag{3.175}$$

where the shorthand is

$$\boxed{\left\{ {d \atop ab} \right\} \coloneqq \frac{1}{2} g^{dc}\left(\frac{\partial}{\partial u^a}\, g_{bc} + \frac{\partial}{\partial u^b}\, g_{ca} - \frac{\partial}{\partial u^c}\, g_{ab}\right)} \tag{3.176}$$

called the *Christoffel symbol of the second kind.*[†6] It is obvious that we have the particular *symmetry* of the solution Γ_{cab} in indices a and b, i.e.,

$$[c, ab] = \partial_{(a}\, g_{b)c} - \frac{1}{2}\,\partial_c\, g_{ab} = [c, ba] \quad \text{and} \quad T_{cba} = 2\,\Gamma_{c[ab]} = 0. \tag{3.177}$$

The solution $[c, ab]$ (or $\left\{ {c \atop ab} \right\}$) is the well-known coefficients of the *Levi–Civita connection*, which indicates that the space considered here is *Riemannian*. Here I would like to remind you that the solution $[c, ab]$ we obtained is implicitly using the conditionds of (i) metricity condition in (3.171), i.e., $\nabla_c\, g_{ab} = 0$; and (ii) the symmetry of $\vec{r}_{ab} = \vec{r}_{ba}$ in (3.172) (this implies $\Gamma_{cab} = \Gamma_{cba}$, which is called the *torsion-free condition*).

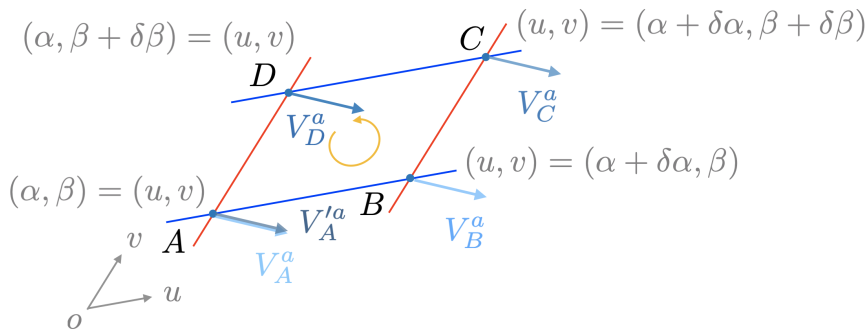

Fig. 3.17: Parallel transport of a vector V^a around a closed loop counterclockwise in a flat space.

[†5] Some people may prefer the notation by $[ab, c]$. In Einstein's paper, you will find that he use the notation $\left[{ab \atop c} \right]$ as the Christoffel symbol of the first kind.

[†6] Einstein use the notation of $\left\{ {ab \atop c} \right\}$ to denote the Christoffel symbol of the second kind.

Curvature tensor. Let's consider a situation that a vector V^a tangent to a *two-dimensional* surface $\mathcal{M}$ is parallel transported around a closed loop *counterclockwise* on $\mathcal{M}$. Cases in flat and curved space are respectively illustrated by Fig. 3.17 and Fig. 3.18. It is obvious that the vector V^a stays the same through the parallel transport if $\mathcal{M}$ is a flat space, i.e., $\mathcal{M} = \mathbb{R}^2$. However, it is different in the case of the curved space. Through the equation of parallel transport (3.148b) or (3.153), we would obtain

$$dV^a = \delta V^a = -\Gamma^a{}_{cu}\, V^c\, du \tag{3.178}$$

if the vector V^a is is parallel transported from A to B along the curve $v = \beta$. By integrating the

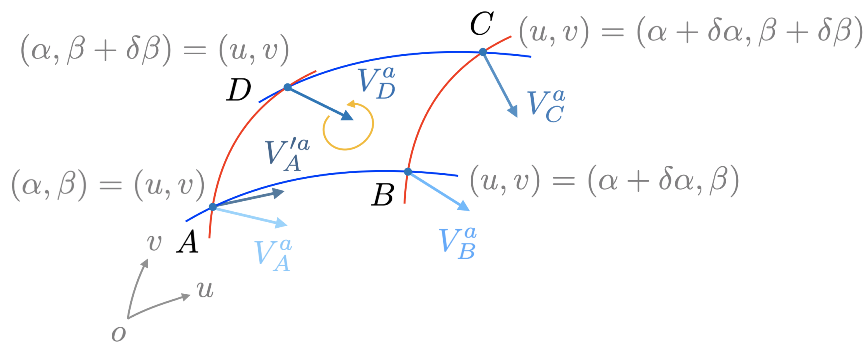

Fig. 3.18: Parallel transport of a vector V^a around a closed loop counterclockwise in a curved space.

equation along u from the point A to point B, the vector V^a at point B should be

$$\begin{aligned} A \to B: \quad V^a_B &= V^a_A + \int_A^B dV^a \\ &= V^a_A + \int_A^B du\, \frac{\partial V^a}{\partial u} \\ &= V^a_A - \int_A^B du\, \left(\Gamma^a{}_{cu}\, V^c\right)_{v=\beta} \\ &= V^a_A - \int_\alpha^{\alpha+\delta\alpha} du\, \left(\Gamma^a{}_{cu}\, V^c\right)_{v=\beta}. \end{aligned} \tag{3.179a}$$

Similarly, we have the integrations along v from the point B to point C, along u from the point C to point D and along v from the point D to point A

$$\begin{aligned} B \to C: \quad V^a_C &= V^a_B - \int_B^C dv\, \left(\Gamma^a{}_{cv}\, V^c\right)_{u=\alpha+\delta\alpha} \\ &= V^a_B - \int_\beta^{\beta+\delta\beta} dv\, \left(\Gamma^a{}_{cv}\, V^c\right)_{u=\alpha+\delta\alpha}, \end{aligned} \tag{3.179b}$$

$$\begin{aligned} C \to D: \quad V^a_D &= V^a_C - \int_C^D d(-u)\, \left(\Gamma^a{}_{cu}\, V^c\right)_{v=\beta+\delta\beta} \\ &= V^a_C + \int_\alpha^{\alpha+\delta\alpha} du\, \left(\Gamma^a{}_{cu}\, V^c\right)_{v=\beta+\delta\beta}, \end{aligned} \tag{3.179c}$$

$$D \to A: \quad V'^a_A = V^a_D - \int_D^A d(-v)\,\left(\Gamma^a{}_{cv}\,V^c\right)_{u\,=\,\alpha}$$

$$= V^a_D + \int_\beta^{\beta+\delta\beta} dv\,\left(\Gamma^a{}_{cv}\,V^c\right)_{u\,=\,\alpha}. \tag{3.179d}$$

As a result, the vector V^a at point A changes and the variation should be

$$\begin{aligned}
\Delta V^a_A &:= V'^a_A - V^a_A \\
&= -\int_\alpha^{\alpha+\delta\alpha} du\,\left(\Gamma^a{}_{cu}\,V^c\right)_{v\,=\,\beta} - \int_\beta^{\beta+\delta\beta} dv\,\left(\Gamma^a{}_{cv}\,V^c\right)_{u\,=\,\alpha+\delta\alpha} \\
&\quad + \int_\alpha^{\alpha+\delta\alpha} du\,\left(\Gamma^a{}_{cu}\,V^c\right)_{v\,=\,\beta+\delta\beta} + \int_\beta^{\beta+\delta\beta} dv\,\left(\Gamma^a{}_{cv}\,V^c\right)_{u\,=\,\alpha} \\
&= +\int_\alpha^{\alpha+\delta\alpha} du\,\Big[\underbrace{\left(\Gamma^a{}_{cu}\,V^c\right)_{v\,=\,\beta+\delta\beta} - \left(\Gamma^a{}_{cu}\,V^c\right)_{v\,=\,\beta}}_{\approx\,\delta\beta\,\frac{\partial}{\partial v}\left(\Gamma^a{}_{cu}\,V^c\right)_{v\,=\,\beta}}\Big] \\
&\quad - \int_\beta^{\beta+\delta\beta} dv\,\Big[\underbrace{\left(\Gamma^a{}_{cv}\,V^c\right)_{u\,=\,\alpha+\delta\alpha} - \left(\Gamma^a{}_{cv}\,V^c\right)_{u\,=\,\alpha}}_{\approx\,\delta\alpha\,\frac{\partial}{\partial u}\left(\Gamma^a{}_{cv}\,V^c\right)_{u\,=\,\alpha}}\Big] \\
&\approx \delta\beta\,\frac{\partial}{\partial v}\left(\Gamma^a{}_{cu}\,V^c\right)_{\substack{u\,=\,\alpha\\ v\,=\,\beta}} \underbrace{\int_\alpha^{\alpha+\delta\alpha} du}_{\delta\alpha} - \delta\alpha\,\frac{\partial}{\partial u}\left(\Gamma^a{}_{cv}\,V^c\right)_{\substack{u\,=\,\alpha\\ v\,=\,\beta}} \underbrace{\int_\beta^{\beta+\delta\beta} dv}_{\delta\beta} \\
&= \delta\alpha\,\delta\beta\left[\frac{\partial}{\partial v}\left(\Gamma^a{}_{cu}\,V^c\right) - \frac{\partial}{\partial u}\left(\Gamma^a{}_{cv}\,V^c\right)\right]_{\substack{u\,=\,\alpha\\ v\,=\,\beta}} \\
&= \delta\alpha\,\delta\beta\Big[\big(\partial_v\Gamma^a{}_{cu}\,V^c + \Gamma^a{}_{cu}\,\underbrace{\partial_v V^c}_{-\Gamma^c{}_{dv}\,V^d}\big) - \big(\partial_u\Gamma^a{}_{cv}\,V^c + \Gamma^a{}_{cv}\,\underbrace{\partial_u V^c}_{-\Gamma^c{}_{du}\,V^d}\big)\Big]_{\substack{u\,=\,\alpha\\ v\,=\,\beta}} \\
&= \delta\alpha\,\delta\beta\Big[\partial_v\Gamma^a{}_{cu} - \partial_u\Gamma^a{}_{cv} + \Gamma^a{}_{dv}\,\Gamma^d{}_{cu} - \Gamma^a{}_{du}\,\Gamma^d{}_{cv}\Big]_A V^c_A,
\end{aligned} \tag{3.180}$$

where $\delta\alpha\,\delta\beta$ is approximately the *area* surrounded by the closed loop. One can find that if the term of the square bracket at point A vanishes, the net change $\Delta V^a_A = 0$, i.e., the vector V'^a_A is parallel to V^a_A. So, such the square bracket term would characterize a geometric quantity and play a crucial role for a space. It is interesting that indices u and v appear antisymmetrically in the square bracket, which implies that the *opposite* transported direction along the closed path leads to different sign of the result.

As a result, the total change of the vector V^a_A is

$$\Delta V^a_A \propto \mathcal{A}^{fb}\Big[\partial_f\Gamma^a{}_{cb} - \partial_b\Gamma^a{}_{cf} + \Gamma^a{}_{df}\,\Gamma^d{}_{cb} - \Gamma^a{}_{db}\,\Gamma^d{}_{cf}\Big]_A V^c_A \tag{3.181}$$

The square bracket is the so-called *Riemann-Christoffel curvature tensor* (or simply the *Riemann curvature tensor* or the *curvature tensor*) at point A

$$\boxed{R^a{}_{cfb}|_A := \Big[\Gamma^a{}_{cb,\,f} - \Gamma^a{}_{cf,\,b} + \Gamma^a{}_{df}\,\Gamma^d{}_{cb} - \Gamma^a{}_{db}\,\Gamma^d{}_{cf}\Big]_A.} \tag{3.182}$$

It is apparent that $R^a{}_{cfb}$ measures the change of the vector V^a at any point within a surface with an area $\mathcal{A}^{fb}$ by

$$\boxed{\Delta V^a \propto \left[R^a{}_{cfb}\right] V^c \mathcal{A}^{fb}.} \tag{3.183}$$

Therefore, if $\Delta V^a \neq 0$ or $R^a{}_{cfb} \neq 0$, it indicates that the space is *curved* as Fig. 3.18, otherwise we have a *flat* space as Fig. 3.17 with $\Delta V^a = 0$ or $R^a{}_{cfb} = 0$. We note that Fig. 3.17 is a specific case with a *vanishing* affinity ($\Gamma^a{}_{bc} = 0$), i.e.,

$$\Gamma^a{}_{bc} = 0 \text{ everywhere} \quad \Longrightarrow \quad R^a{}_{cfb} = 0 \text{ everywhere.} \tag{3.184}$$

Flat space

We call a space *flat* if and only if the curvature tensor *vanishes everywhere*, i.e., $R^a{}_{cfb}(u, v) = 0$ *everywhere*.

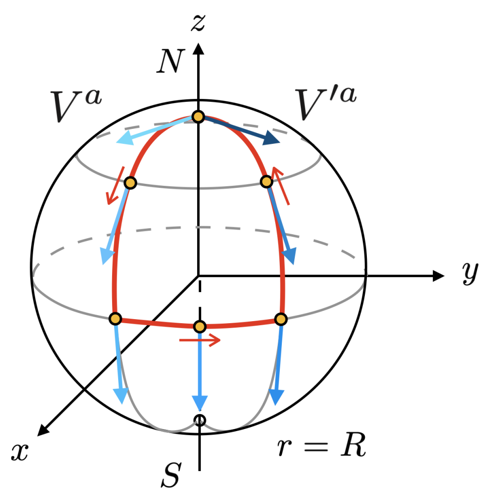

Fig. 3.19: Parallelly transported a vector V^a along a closed loop (red solid line) counterclockwise on a sphere.

A well-constructed example of the effect of the existence of the curvature can be illustrated by a sphere with coordinates $\{u^a\} = \{\theta, \varphi\}$, we focus on a particular tangent vector V^a at its North pole, N. Parallelly transported the vector along the path *counterclockwise* by keeping the directional cosine constant, and finally returned the North pole, N, which is shown in Fig. 3.19. It is apparent that the transported vector V'^a is not parallel to itself before the parallel transport.

Note: $\Gamma^a{}_{bc} = 0$ is *not* a condition to determine a flat space. In other words, we can have a *non-vanishing* affinity in the flat space.

3.8 Differential Relations

If we consider to apply the covariant differentiation on a scalar field *twice*

$$\boldsymbol{D}_c \boldsymbol{D}_a \phi = \boldsymbol{D}_c \partial_a \phi = \partial_c \partial_a \phi. \tag{3.185a}$$

So it is soon to obtian the commutator

$$\boxed{\big(\boldsymbol{D}_c\boldsymbol{D}_a - \boldsymbol{D}_a\boldsymbol{D}_c\big)\phi = 0}. \tag{3.185b}$$

It can be directly check the result of $\nabla_c\nabla_a\phi$ by expansion

$$\nabla_c\nabla_a\phi = \phi_{;\,ac} = \big(\phi_{;\,a}\big)_{;\,c} = \underbrace{\big(\nabla_a\phi\big)_{,\,c}}_{(\phi_{,\,a})_{,\,c}} - \big(\nabla_b\phi\big)\Gamma^b{}_{ac} = \phi_{,\,ac} - \phi_{,\,b}\,\Gamma^b{}_{ac}. \tag{3.185c}$$

We note that ∇ or the *semicolon* ";" in (3.185c) is a short shorthand to denoted two terms (the derivative and connection terms), if we need to use these notation to do the calculation, we need to do the *expansion* on the equation *from outer to inner* since the connection is not a tensor and we do not know how to do the covariant derivative on it. By exchange the indices a and c, we can easily have a similar result:

$$\nabla_a\nabla_c\phi = \phi_{,\,ca} - \phi_{,\,b}\,\Gamma^b{}_{ca}. \tag{3.186}$$

So we obtain an identity

$$\boxed{\big(\nabla_c\nabla_a - \nabla_a\nabla_c\big)\phi = \phi_{,\,b}\big(\Gamma^b{}_{ca} - \Gamma^b{}_{ac}\big) = \phi_{,\,b}\,T^b{}_{ac}.} \tag{3.187}$$

Again, for *Levi–Civita connection*, the right hand side of the identity is vanished:

$$\boxed{\big(\nabla_c\nabla_a - \nabla_a\nabla_c\big)\phi = 0.} \tag{3.188}$$

Apply the covariant derivative twice on $\vec{V}$ is

$$\begin{aligned}
\boldsymbol{D}_b\boldsymbol{D}_c\vec{V} &= \boldsymbol{D}_b\boldsymbol{D}_c\big(e_a\,V^a\big) = \boldsymbol{D}_b\big[(\underbrace{\boldsymbol{D}_c e_a}_{e_d\,\Gamma^d{}_{ac}})V^a + e_a\,V^a{}_{,\,c}\big] = \boldsymbol{D}_b\big[(e_d\,\Gamma^d{}_{ac})V^a + e_a\,V^a{}_{,\,c}\big] \\
&= \big[(\underbrace{\boldsymbol{D}_b e_d}_{e_e\,\Gamma^e{}_{db}})\,\Gamma^d{}_{ac} + e_d\,\Gamma^d{}_{ac,\,b}\big]V^a + (e_b\,\Gamma^b{}_{ac})\,V^a{}_{,\,b} + e_e\,\Gamma^e{}_{ab}\,V^a{}_{,\,c} + e_a\,V^a{}_{,\,cb}\big] \\
&= e_a\Big\{\big(\Gamma^a{}_{db}\,\Gamma^d{}_{ec} + \Gamma^a{}_{ec,\,b}\big)V^a + \Gamma^a{}_{ec}\,V^e{}_{,\,b} + \Gamma^a{}_{eb}\,V^e{}_{,\,c} + V^a{}_{,\,cb}\Big\},
\end{aligned} \tag{3.189}$$

and

$$\boldsymbol{D}_c\boldsymbol{D}_b\vec{V} = e_a\Big\{\big(\Gamma^a{}_{dc}\,\Gamma^d{}_{eb} + \Gamma^a{}_{eb,\,c}\big)V^a + \Gamma^a{}_{eb}\,V^e{}_{,\,c} + \Gamma^a{}_{ec}\,V^e{}_{,\,b} + V^a{}_{,\,bc}\Big\}. \tag{3.190}$$

So we have

$$\boxed{\big(\boldsymbol{D}_b\boldsymbol{D}_c - \boldsymbol{D}_c\boldsymbol{D}_b\big)\vec{V} = e_a\,R^a{}_{ebc}\,V^e.} \tag{3.191}$$

It is interesting to apply ∇ twice on the *contravariant* component of $\vec{V}$

$$\begin{aligned}
\nabla_b\nabla_c V^a &= V^a{}_{;\,cb} = \big(V^a{}_{;\,c}\big)_{;\,b} \\
&= \big(V^a{}_{;\,c}\big)_{,\,b} + \Gamma^a{}_{db}\,V^d{}_{;\,c} - V^a{}_{;\,d}\,\Gamma^d{}_{cb} \\
&= \big(V^a{}_{,\,c} + \Gamma^a{}_{ec}\,V^e\big)_{,\,b} + \Gamma^a{}_{db}\,\big(V^d{}_{,\,c} + \Gamma^d{}_{ec}\,V^e\big) - \big(V^a{}_{,\,d} + \Gamma^a{}_{ed}\,V^e\big)\,\Gamma^d{}_{cb} \\
&= V^a{}_{,\,cb} + \Gamma^a{}_{ec,\,b}\,V^e + \Gamma^a{}_{ec}\,V^e{}_{,\,b} + \Gamma^a{}_{db}\,V^d{}_{,\,c} + \Gamma^a{}_{db}\,\Gamma^d{}_{ec}\,V^e
\end{aligned}$$

$$
\begin{aligned}
&- V^a{}_{,d}\,\Gamma^d{}_{cb} - \Gamma^a{}_{ed}\,V^e\,\Gamma^d{}_{cb} \\
&= V^a{}_{,cb} + \Gamma^a{}_{ec}\,V^e{}_{,b} + \Gamma^a{}_{db}\,V^d{}_{,c} - V^a{}_{,d}\,\Gamma^d{}_{cb} \\
&\quad + \left(\Gamma^a{}_{ec,b} + \Gamma^a{}_{db}\,\Gamma^d{}_{ec} - \Gamma^a{}_{ed}\,\Gamma^d{}_{cb}\right) V^e,
\end{aligned}
\tag{3.192}
$$

and

$$
\begin{aligned}
\nabla_c \nabla_b V^a &= V^a{}_{,bc} + \Gamma^a{}_{eb}\,V^e{}_{,c} + \Gamma^a{}_{dc}\,V^d{}_{,b} - V^a{}_{,d}\,\Gamma^d{}_{bc} \\
&\quad + \left(\Gamma^a{}_{eb,c} + \Gamma^a{}_{dc}\,\Gamma^d{}_{eb} - \Gamma^a{}_{ed}\,\Gamma^d{}_{bc}\right) V^e,
\end{aligned}
\tag{3.193}
$$

which results in the well-known *Ricci identity*

$$
\boxed{(\nabla_b \nabla_c - \nabla_c \nabla_b) V^a = V^a{}_{;cb} - V^a{}_{;bc} = R^a{}_{ebc}\,V^e - V^a{}_{;d}\,T^d{}_{bc}.}
\tag{3.194}
$$

Again, for *Levi–Civita connection*, the tensor $T^d{}_{bc}$ should be vanished, and the Ricci identity becomes

$$
\boxed{(\nabla_b \nabla_c - \nabla_c \nabla_b) V^a = R^a{}_{ebc}\,V^e.}
\tag{3.195}
$$

3.9 Weyl Tensor

The trace-free part of the Riemann tensor is defined as the *Weyl tensor*. For n-dimensional space, the Weyl tensor is expressed by

$$
C^{ab}{}_{cd} = R^{ab}{}_{cd} - \frac{4}{n-2}\,\delta^{[a}{}_{[c}\,R^{b]}{}_{d]} + \frac{2}{(n-1)(n-2)}\,\delta^{[a}{}_{[c}\,\delta^{b]}{}_{d]}\,R.
\tag{3.196}
$$

For $n = 4$ dimensions, it reads as

$$
\begin{aligned}
C^{ab}{}_{cd} &= R^{ab}{}_{cd} - 2\,\delta^{[a}{}_{[c}\,R^{b]}{}_{d]} + \frac{1}{3}\,\delta^{[a}{}_{[c}\,\delta^{b]}{}_{d]}\,R \\
&= R^{ab}{}_{cd} - \frac{1}{2}\left(\delta^a{}_b\,R^c{}_d - \delta^a{}_b\,R^c{}_d + \delta^c{}_d\,R^a{}_b - \delta^c{}_b\,R^a{}_d\right) \\
&\quad + \frac{1}{6}\left(\delta^a{}_c\,\delta^b{}_d - \delta^a{}_d\,\delta^b{}_c\right) R.
\end{aligned}
\tag{3.197}
$$

It is apparent that if $R_{\mu\nu} = 0$, $R = 0$ is automatically satisfied, then it results in that the Riemann tensor is fully determined by the Weyl tensor, i.e., $R^{ab}{}_{cd} = C^{ab}{}_{cd}$. Consider a transformation of the line element or the metric *scaling* by a non-negative function $\Omega^2(x)$, called the *conformal transformation*:[†7]

$$
ds^2 \longrightarrow \widetilde{ds}^2 = \Omega^2(x)\,ds^2 \quad \Longrightarrow \quad g_{ab} \longrightarrow \widetilde{g}_{ab} = \Omega^2(x)\,g_{ab}.
\tag{3.198}
$$

Hence, it preserves the angle between any two basis vectors. Given a vector $\vec{V}$, such a transformation merely changes the norm of the vector

$$
\|\vec{V}\|^2 = g(\vec{V},\vec{V}) = g_{ab}\,V^a\,V^b \longrightarrow \|\vec{V}\|^2 = \widetilde{g}(\vec{V},\vec{V}) = \Omega^2(x)\,g_{ab}\,V^a\,V^b.
\tag{3.199}
$$

It can be shown that, the Weyl tensor is *invariant* under the conformal transformation, i.e.,

$$
\boxed{C^{ab}{}_{cd} \longrightarrow \widetilde{C}^{ab}{}_{cd} = C^{ab}{}_{cd}.}
\tag{3.200}
$$

Because the invariance property under the conformal transformation, the Weyl tensor is also called Weyl's *conformal tensor*.

[†7] We note that this is different from the conformal transformation in the *conformal field theory*, which is a coordinate transformation. In order to distinguish the latter, some people call the former the *Weyl rescaling*

3.10 Gauss-Codazzi equation

Now we have Gauss and Weingarten formulas

$$\begin{cases} \vec{r}_{ac} = \vec{r}_d \, \Gamma^d{}_{ac} + \vec{n} \, b_{ac}, & (3.201a) \\ \vec{n}_b = -\vec{r}_d \, b^d{}_b, & (3.201b) \end{cases}$$

which correspond to the derivative vectors of tangent and normal vectors respectively. By taking the partial derivative of $\vec{r}_{ac}$ with respect to u^b, we have

$$\partial_b \vec{r}_{ac} = \vec{r}_d \big(\partial_b \Gamma^d{}_{ac} + \Gamma^d{}_{eb} \Gamma^e{}_{ac} - b^d{}_b b_{ac} \big) + \vec{n} \big(b_{db} \Gamma^d{}_{ac} + \partial_b b_{ac} \big). \tag{3.202}$$

By interchanging the indices b and c, we obtain

$$\partial_c \vec{r}_{ab} = \vec{r}_d \big(\partial_c \Gamma^d{}_{ab} + \Gamma^d{}_{ec} \Gamma^e{}_{ab} - b^d{}_c b_{ab} \big) + \vec{n} \big(b_{dc} \Gamma^d{}_{ab} + \partial_c b_{ab} \big). \tag{3.203}$$

We note that

$$\partial_b \vec{r}_{ac} - \partial_c \vec{r}_{ab} = \partial_b \partial_c \vec{r}_a - \partial_c \partial_b \vec{r}_a = 0. \tag{3.204}$$

According to (3.202), (3.203) and (3.204), we obtain a couple of equations called *Gauss-Codazzi equation*

$$\begin{aligned} 0 &= \partial_b \vec{r}_{ac} - \partial_c \vec{r}_{ab} \\ &= \vec{r}_d \big(\underbrace{R^d{}_{abc} - b^d{}_b b_{ac} + b^d{}_c b_{ab}}_{0} \big) + \vec{n} \big(\underbrace{b_{db} \Gamma^d{}_{ac} - b_{dc} \Gamma^d{}_{ab} + \partial_b b_{ac} - \partial_c b_{ab}}_{0} \big) \end{aligned} \tag{3.205}$$

due to linearity of $\{\vec{r}_d, \vec{n}\}$, which are usually written as

$$\begin{cases} \textit{Gauss equation:} \ \boxed{R^d{}_{abc} = b^d{}_b b_{ac} - b^d{}_c b_{ab},} & (3.206a) \\ \textit{Codazzi equation:} \ \boxed{\partial_b b_{ac} - \partial_c b_{ab} = b_{dc} \Gamma^d{}_{ab} - b_{db} \Gamma^d{}_{ac}.} & (3.206b) \end{cases}$$

Here the Riemann(-Christoffel) curvature tensor can also be obtained

$$\boxed{R^d{}_{abc} \coloneqq \partial_b \Gamma^d{}_{ac} - \partial_c \Gamma^d{}_{ab} + \Gamma^d{}_{eb} \Gamma^e{}_{ac} - \Gamma^d{}_{ec} \Gamma^e{}_{ab}.} \tag{3.207}$$

Using the metric tensor, the four lowing-indices curvature tensor can be defined by

$$\boxed{R_{dabc} \coloneqq g_{de} R^e{}_{abc}.} \tag{3.208}$$

We note some symmetries of the Riemann curvature tensor here:

- Antisymmetric in the indices b and c

$$R^d{}_{acb} = -R^d{}_{abc}. \tag{3.209a}$$

- Antisymmetric in d and a (*only* in Riemann space)

$$R_{adbc} = -R_{dabc}. \tag{3.209b}$$

- Symmetric in the pairs of indices (da) and (bc) (*only* in Riemannian space)

$$R_{bcda} = +R_{dabc}. \tag{3.209c}$$

- The cyclic relation (*first Bianchi identity*)

$$R^d{}_{acb} + R^d{}_{cba} + R^d{}_{bac} = 0 \quad \text{or} \quad R^d{}_{[acb]} = 0. \tag{3.209d}$$

We can also contract the first and the third indices in curvature tensor called *Ricci tensor*, which is given by

$$\boxed{R_{ac} = R^b{}_{abc} = g^{db} R_{dabc},} \tag{3.210}$$

and then the *scalar curvature* or *Ricci scalar* is defined by

$$\boxed{\mathbf{R} = R^c{}_c = g^{ac} R_{ac} \overset{\text{or}}{=} g^{ac} g^{db} R_{dabc}.} \tag{3.211}$$

We note the symmetry of the Ricci curvature tensor here:

$$R_{ca} = +R_{ac}. \tag{3.212}$$

3.11 Theorema Egregium of Gauss

For *two*-dimensional surface $\mathcal{M}$ in $\mathbb{R}^3$, the indices $a, b, c, d = 1, 2$, by the properties of Riemann curvature tensor, the following components are vanished:

$$R_{11bc} = R_{22bc} = 0\,, \quad R_{da11} = R_{da22} = 0\,, \tag{3.213}$$

From the Gauss equation (3.206a), the residual one component should uniquely be

$$\begin{aligned} R_{1212} &= b_{11} b_{22} - b_{12} b_{12} \\ &= LN - M^2 \\ &= \det(b_{ab}) \coloneqq \mathbf{b}\,. \end{aligned} \tag{3.214}$$

Therefore, we can rewrite the Gauss curvature (3.120) as

$$\boxed{K = \frac{\mathbf{b}}{\mathbf{g}} = \frac{R_{1212}}{\mathbf{g}}\,,} \tag{3.215}$$

which is a function of g_{ab} *only*, i.e., a 2-dimensional surface in $\mathbb{R}^3$ is *totally* determined by it's intrinsic structure associated with g_{ab}. This is the famous *intrinsic geometry of Gauss* and we call this the *theorema egregium of Gauss*. From the Codazzi equation (3.206b), we only need to consider the case of $a = 1, 2$ and $b = 1$ as well as $c = 2$. This gives

$$\begin{cases} \dfrac{\partial b_{12}}{\partial u^1} - \dfrac{\partial b_{11}}{\partial u^2} = b_{d2} \Gamma^d{}_{11} - b_{d1} \Gamma^d{}_{12}\,, & \qquad (3.216\text{a}) \\[2ex] \dfrac{\partial b_{22}}{\partial u^1} - \dfrac{\partial b_{21}}{\partial u^2} = b_{d2} \Gamma^d{}_{21} - b_{d1} \Gamma^d{}_{22}\,, & \qquad (3.216\text{b}) \end{cases} \quad (d = 1, 2)\,.$$

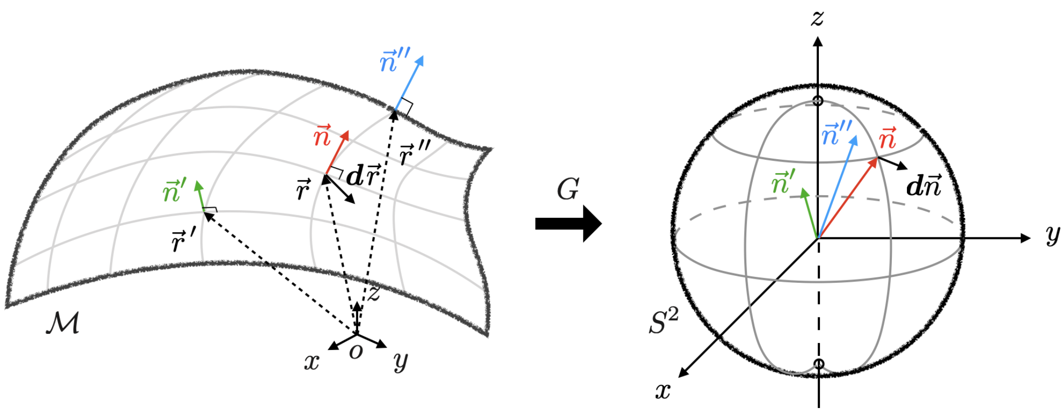

Fig. 3.20: The Gauss map G of a surface.

Third fundamental (quadratic) form. In analogy, we have a Gauss sphere S^2 is shown in Fig. 3.20, which is the image of the Gauss map G from a two-dimensional surface $\mathcal{M}$. A normal vector $\vec{n}$ will be sent to be a radius vector of S^2. Therefore, like the conclusion of (3.22) for a curve in $\mathbb{R}^2$, $\vec{n}$ on S^2 play the same role as $\vec{r}$ on the surface $\mathcal{M}$. As a result, we can calculate the first fundamental for of S^2:

$$\mathrm{I}|_{\mathcal{M}} = d\vec{r} \cdot d\vec{r} \quad \longrightarrow \quad \mathrm{I}|_{S^2} = d\vec{n} \cdot d\vec{n}. \tag{3.217}$$

Now we define the *third fundamental form* of $\mathcal{M}$ to be the first fundamental form of the Gauss sphere S^2

$$\mathrm{III}|_{\mathcal{M}} := \mathrm{I}|_{S^2} = d\vec{n} \cdot d\vec{n}. \tag{3.218}$$

From the first fundamental form, we can calculate the area element on the surface $\mathcal{M}$ and S^2, which are denoted by $d\mathcal{A}|_{\mathcal{M}}$ and $d\mathcal{A}|_{S^2}$ respectively. The results can be obtained by

$$\begin{cases} d\vec{r} \approx \vec{r}_u\, du + \vec{r}_v\, dv \quad \Longrightarrow \quad d\mathcal{A}|_{\mathcal{M}} = \|\vec{r}_u\, du \wedge \vec{r}_v\, dv\| = \|\vec{r}_u \wedge \vec{r}_v\|\, du\, dv, & \text{(3.219a)} \\ d\vec{n} \approx \vec{n}_u\, du + \vec{n}_v\, dv \quad \Longrightarrow \quad d\mathcal{A}|_{S^2} = \|\vec{n}_u\, du \wedge \vec{n}_v\, dv\| = \|\vec{n}_u \wedge \vec{n}_v\|\, du\, dv. & \text{(3.219b)} \end{cases}$$

However, the vectors $\vec{n}_u$ and $\vec{n}_v$ are given by the Weingarten formulas (3.100) with (3.101) and (3.102). We can express $\vec{n}_u \wedge \vec{n}_v$ in terms of $\vec{r}_u$ and $\vec{r}_v$ by

$$\begin{aligned} \vec{n}_u \wedge \vec{n}_v &= AD(\vec{r}_u \wedge \vec{r}_v) - BC(\vec{r}_u \wedge \vec{r}_v) \\ &= \left(\frac{FM - GL}{EG - F^2}\right)\left(\frac{FM - EN}{EG - F^2}\right)(\vec{r}_u \wedge \vec{r}_v) - \left(\frac{FL - EM}{EG - F^2}\right)\left(\frac{FN - GM}{EG - F^2}\right)(\vec{r}_u \wedge \vec{r}_v) \\ &= \frac{LN - M^2}{EG - F^2}(\vec{r}_u \wedge \vec{r}_v) = \frac{\mathbf{b}}{\mathbf{g}}(\vec{r}_u \wedge \vec{r}_v) \underset{\substack{\uparrow \\ \text{By (3.120)}}}{=} K(\vec{r}_u \wedge \vec{r}_v)\,. \end{aligned} \tag{3.220}$$

Consequently, cf. (3.27), we can measure the absolute value of the Gauss curvature by the ratio

$$\frac{d\mathcal{A}|_{S^2}}{d\mathcal{A}|_{\mathcal{M}}} = \frac{\|\vec{n}_u \wedge \vec{n}_v\|\, du\, dv}{\|\vec{r}_u \wedge \vec{r}_v\|\, du\, dv} = \frac{|K|\|\vec{r}_u \wedge \vec{r}_v\|\, du\, dv}{\|\vec{r}_u \wedge \vec{r}_v\|\, du\, dv} = |K|\,. \tag{3.221}$$

Appendix

A Curvilinear Coordinates

In many cases it is not convenient to use to Cartesian coordinates for calculations. We review the so-called *curvilinear coordinates* here.

Polar coordinates in $\mathbb{R}^2$

- The displacement vector:

$$\boldsymbol{d}\vec{r} = \vec{e}_r\, dr + \vec{e}_\theta (r\, d\theta). \tag{A.1a}$$

- Vectorial variations:

$$\boldsymbol{d}\vec{e}_r = \vec{e}_\theta\, d\theta, \quad \boldsymbol{d}\vec{e}_\theta = -\, \vec{e}_r\, d\theta. \tag{A.1b}$$

- The line element:

$$dS^2 = dr^2 + r^2\, d\theta^2. \tag{A.1c}$$

- The surface element:

$$\begin{aligned} d\mathcal{A} &= d\ell_{\text{circle}}|_r \cdot dr \\ &= (r\, d\theta) dr = r\, dr\, d\theta. \end{aligned} \tag{A.1d}$$

Cylindrical coordinates in $\mathbb{R}^3$

- The displacement vector:

$$\boldsymbol{d}\vec{r} = \vec{e}_\rho\, d\rho + \vec{e}_\varphi (\rho\, d\varphi) + \vec{e}_z\, dz. \tag{A.2a}$$

- Vectorial variations:

$$\boldsymbol{d}\vec{e}_\rho = \vec{e}_\varphi\, d\varphi, \quad \boldsymbol{d}\vec{e}_\varphi = -\, \vec{e}_\rho\, d\varphi, \quad \boldsymbol{d}\vec{e}_z = 0. \tag{A.2b}$$

- The line element:

$$dS^2 = d\rho^2 + \rho^2\, d\varphi^2 + dz^2. \tag{A.2c}$$

- The surface element vector:

$$\overrightarrow{d\mathcal{A}} = \vec{e}_\rho\, d\mathcal{A}|_\rho + \vec{e}_\varphi\, d\mathcal{A}|_\varphi + \vec{e}_z\, d\mathcal{A}|_z \tag{A.2d}$$

$$= \vec{e}_\rho\, (\rho\, d\varphi) dz + \vec{e}_\varphi\, d\rho\, dz + \vec{e}_z\, d\rho (\rho\, d\varphi). \tag{A.2e}$$

- The volume element:

$$\begin{aligned} d\mathcal{V} &= d\mathcal{A}_{\text{cylinder}}|_\rho \cdot d\rho \\ &= \big[(\rho\ d\varphi) dz\big] d\rho = \rho\, d\rho\, d\varphi\, dz. \end{aligned} \tag{A.2f}$$

Spherical coordinates in $\mathbb{R}^3$

- The displacement vector:

$$\boldsymbol{d}\vec{r} = \vec{e}_r\, dr + \vec{e}_\theta (r\, d\theta) + \vec{e}_\varphi (r \sin\theta\, d\varphi). \tag{A.3a}$$

- Vectorial variations:

$$\begin{aligned}
&\boldsymbol{d}\vec{e}_r = \vec{e}_\theta\, d\theta + \vec{e}_\varphi\, (\sin\theta\, d\varphi), \quad \boldsymbol{d}\vec{e}_\theta = -\vec{e}_r\, d\theta, \\
&\boldsymbol{d}\vec{e}_\varphi = -\vec{e}_r \left(\frac{d\varphi}{\sin\theta}\right) + \vec{e}_z(\cot\theta\, d\varphi) = -\vec{e}_r \left(\frac{d\varphi}{\sin\theta}\right) + \frac{1}{2}\big(\vec{e}_r \cos\theta - \vec{e}_\theta \sin\theta\big)(\cot\theta\, d\varphi).
\end{aligned} \tag{A.3b}$$

- The line element:

$$dS^2 = dr^2 + r^2\, d\theta^2 + r^2 \sin^2\theta\, d\varphi^2. \tag{A.3c}$$

- The surface element vector:

$$\overrightarrow{d\mathcal{A}} = \vec{e}_r\, d\mathcal{A}|_r + \vec{e}_\theta\, d\mathcal{A}|_\theta + \vec{e}_\varphi\, d\mathcal{A}|_\varphi \tag{A.3d}$$

$$= \vec{e}_r\, (r\, d\theta)(\sin\theta\, d\varphi) + \vec{e}_\theta\, dr (r\, d\theta) + \vec{e}_\varphi\, dr (r \sin\theta\, d\varphi). \tag{A.3e}$$

- The volume element:

$$\begin{aligned}
d\mathcal{V} &= d\mathcal{A}_{\text{sphere}}|_r \cdot dr \\
&= \big[(r\, d\theta)(r \sin\theta\, d\varphi)\big] dr = r^2 \sin\theta\, dr\, d\theta\, d\varphi.
\end{aligned} \tag{A.3f}$$